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1. Introduction

This document describes a GitHub App that reviews pull requests using AWS Bedrock and Lambda. The authored functional architecture remains stable while deployment/runtime/code evidence is inferred from infrastructure manifests, source files, and verification artifacts.

1.1. Scope and Assumptions

- **Functional Groups** and **Functional Units** are the stable authored design backbone.
- **Runtime elements** and **code elements** are inferred from available IaC, deployment, and source artifacts.
- Inferred coverage is evidence-driven and may be partial when source metadata is missing.
- **Requirements** and reference indexes are provided for traceability, not as primary narrative flow.

2. How To Read This Document

This document is intentionally split into two layers:

1. Concept layer: functional and view-specific architecture narratives for human understanding.
2. Evidence layer: generated requirement traceability and inferred reference indexes.

Read in this order:

- 1. Architecture Intent View: stable authored architecture intent and ownership boundaries.
- 2. Communication / Deployment / Security Views: viewpoint-specific operational behavior.
- 3. Traceability View: requirement mapping, evidence coverage, and confidence.
- 4. Reference Appendix: supporting reference IDs and inferred-source links.

3. Document Health Snapshot

View	Authored Status	Authored Status Explanation	Units in Scope	Direct Evidence Coverage	Gap Count	Heuristic Confidence	Heuristic Basis
Architecture Intent View (VIEW-ARCHITECTURE-INTENT)	stable	Authored capability boundaries and ownership responsibilities are stable for this release.	7	7	0	high	Defaults authored units to covered; use authoredStatus as the primary publication signal.
Communication View (VIEW-COMMUNICATION)	in-review	Communication sequencing is defined, while protocol-level metadata is still being expanded.	7	3	4	low	Counts a unit as covered when direct inferred runtime evidence exists.
Deployment View (VIEW-DEPLOYMENT)	in-review	Deployment and runtime mapping are present; observability and ownership evidence are still evolving.	7	3	4	low	Counts a unit as covered when direct inferred runtime evidence exists.
Security View (VIEW-SECURITY)	draft	Core threats are modeled; control-depth and detection mappings are still being expanded.	7	7	2	high	Counts a unit as covered when explicit authored attack-vector mapping exists.

View	Authored Status	Authored Status Explanation	Units in Scope	Direct Evidence Coverage	Gap Count	Heuristic Confidence	Heuristic Basis
Traceability View (VIEW-TRACEABILITY)	in-review	Requirement mappings exist; evidence completeness and consistency are still being hardened.	7	7	4	high	Counts a unit as covered when direct inferred runtime or code evidence exists.
State Lifecycle View (VIEW-STATE-LIFECYCLE)	in-review	Review lifecycle states and guarded transitions are explicitly modeled.	0	0	0	n/a	Lifecycle heuristic is currently basic; treat authoredStatus as primary signal.
Interaction Flow View (VIEW-INTERACTION-FLOW)	in-review	Interaction flow captures PR review causality from ingress through publication.	4	4	0	high	Flow heuristic is step/branch-oriented; treat authoredStatus plus projection completeness as the primary signal.

4. Terms and Definitions

Term	Definition
actor (EM-ACTOR)	A person or operational role that interacts with functional units.
ai review is enabled (FEAT-AI-REVIEW)	Feature flag enabling model-assisted review. Aliases:ai review enabled
attack vector (EM-ATTACK-VECTOR)	A technical misuse or attack path that targets functional, referenced, or runtime elements.
audit logging is enabled (FEAT-AUDIT-LOGGING)	Feature flag enabling audit metadata persistence. Aliases:audit enabled
authored mapping (EM-AUTHORED-MAPPING)	An explicit relationship declared in architecture inputs between authored or referenced elements.
aws sdk for go (REF-AWS-SDK-GO)	SDK dependency used for AWS runtime integrations. Aliases:aws sdk
aws secrets manager (REF-AWS-SECRETS-MANAGER)	Managed secret store for application credentials and config values. Aliases:secrets manager
bedrock runtime (REF-AWS-BEDROCK-RUNTIME)	External Bedrock runtime endpoint used for model inference. Aliases:bedrock api

Term	Definition
bedrock runtime is unavailable (EVT-BEDROCK-UNAVAILABLE)	Event raised when Bedrock inference path is degraded. Aliases:bedrock unavailable
check run summary (DATA-CHECKRUN-SUMMARY)	Aggregated review result posted as a GitHub check run. Aliases:check run summaries
code element (EM-CODE-ELEMENT)	An inferred code structure or ownership element discovered from source trees and build metadata.
control (EM-CONTROL)	An authored security, policy, or operational control used to constrain behavior and reduce risk.
data object (EM-DATA-OBJECT)	An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.
deployment target (EM-DEPLOYMENT-TARGET)	An authored deployment destination such as environment, cluster, namespace, or registry scope.
developer (ACT-DEVELOPER)	Engineer opening and updating pull requests. Aliases:pr author
diff context (DATA-DIFF-CONTEXT)	Pull request file changes and metadata prepared for analysis. Aliases:normalized review context, pull request files, pull request metadata, review context, review-ready context
diff context is prepared (EVT-CONTEXT-PREPARED)	Event indicating context assembly completed and is ready for analysis. Aliases:context prepared
event (EM-EVENT)	An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.
flow (EM-FLOW)	An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.
flow step (EM-FLOW-STEP)	A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.
functional group (EM-FUNCTIONAL-GROUP)	A major authored capability area that groups related functional units.
functional unit (EM-FUNCTIONAL-UNIT)	An authored working unit inside a functional group that owns specific behavior.
github app api (REF-GITHUB-APP-API)	External GitHub API used for pull request metadata and review publication. Aliases:github api
github rest sdk (REF-GITHUB-REST-SDK)	SDK dependency used for GitHub App REST interactions. Aliases:github sdk

Term	Definition
github webhook ingress (FU-GITHUB-WEBHOOK-INGRESS)	Validates webhook signatures and routes trusted pull request events. Aliases:webhook ingress
inferred mapping (EM-INFERRED-MAPPING)	A discovered relationship that links inferred runtime/code elements upward to authored design.
interface (EM-INTERFACE)	An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.
lambda runtime operations (FU-LAMBDA-RUNTIME-OPERATIONS)	Manages runtime deployment and operational controls. Aliases:lambda ops
normal mode is active (MODE-NORMAL)	Default operating mode using AI plus deterministic checks. Aliases:normal mode
platform operations (FG-PLATFORM-OPERATIONS)	Functional group for deployment lifecycle and secret/config controls. Aliases:runtime operations
platform operator (ACT-PLATFORM-OPERATOR)	Owner of deployment/runtime operations and reliability controls. Aliases:platform ops
policy checks (FU-POLICY-CHECKS)	Evaluates deterministic repository policy checks. Aliases:deterministic checks
policy checks are enabled (FEAT-POLICY-CHECKS)	Feature flag enabling deterministic policy checks. Aliases:policy checks enabled
policy ruleset (REF-POLICY-RULESET)	Deterministic review policy bundle used during analysis. Aliases:policy bundle
pr context assembly (FU-PR-CONTEXT-ASSEMBLY)	Fetches changed files and prepares bounded review context payloads. Aliases:context assembly
pr integration (FG-PR-INTEGRATION)	Functional group for webhook ingress and pull request context intake. Aliases:github integration
pr review system (SYS-PR-REVIEW-SYSTEM)	GitHub App review system-of-interest that ingests PR events and publishes review outcomes. Aliases:the pr review system
prompt injection in diff (AV-PROMPT-INJECTION-IN-DIFF)	Malicious prompt-like content embedded in pull request code changes. Aliases:prompt injection
pull request opened event is received (EVT-PR-OPENED)	Event raised when a new pull request is opened. Aliases:pr opened, webhook events
pull request review is active (STATE-PR-REVIEW-ACTIVE)	State where review processing is currently active for a pull request. Aliases:review active

Term	Definition
pull request spam abuse (AV-PR-SPAM-ABUSE)	High-volume event abuse causing review capacity exhaustion. Aliases:pr spam
pull request synchronize event is received (EVT-PR-SYNCHRONIZE)	Event raised when new commits are pushed to an open pull request. Aliases:authorized events, pr synchronize
redacted audit metadata (DATA-REDACTED-AUDIT-METADATA)	Sanitized prompt and finding metadata retained for audit.
referenced element (EM-REFERENCED-ELEMENT)	An architecture-relevant external, platform, or third-party dependency represented by role.
repository maintainer (ACT-REPOSITORY-MAINTAINER)	Owner of merge decisions and repository quality gates. Aliases:maintainer
requirement (EM-REQUIREMENT)	A structured requirement statement that defines expected system behavior and maps to functional ownership.
result metadata (DATA-RESULT-METADATA)	Metadata describing review output characteristics for audit and reporting.
review findings are generated (EVT-FINDINGS-GENERATED)	Event indicating analysis output is ready for publication. Aliases:findings generated, review outcome
review findings are published (EVT-FINDINGS-PUBLISHED)	Event indicating review output has been published to GitHub checks/comments. Aliases:findings published, publication output
review intelligence (FG-REVIEW-INTELLIGENCE)	Functional group for AI and deterministic review analysis and publication. Aliases:review system
review orchestration (FU-REVIEW-ORCHESTRATION)	Coordinates Bedrock analysis with deterministic checks and fallback logic. Aliases:orchestration
review publication (FU-REVIEW-PUBLICATION)	Publishes review findings to GitHub checks and comments. Aliases:publication
review request rate limit is exceeded (EVT-RATE-LIMIT-EXCEEDED)	Event raised when PR review rate guardrails are hit. Aliases:rate limit exceeded
runtime element (EM-RUNTIME-ELEMENT)	An inferred runtime realization element discovered from infrastructure and deployment sources.
secret leakage in review output (AV-SECRET-LEAKAGE-IN-REVIEW)	Unintended credential disclosure in generated review findings. Aliases:secret leakage
secrets and configuration (FU-SECRETS-CONFIGURATION)	Manages credentials and secure runtime configuration. Aliases:secret config

Term	Definition
security engineer (ACT-SECURITY-ENGINEER)	Owner of security policy checks and finding triage. Aliases:security reviewer
spoofed github webhook (AV-SPOOFED-GITHUB-WEBHOOK)	Forged webhook request attempting unauthorized review execution. Aliases:webhook spoofing
state (EM-STATE)	An authored lifecycle state used to model transition behavior, guards, and event-driven progression.
trust boundary (EM-TRUST-BOUNDARY)	An authored trust separation zone that marks policy, identity, or security control boundaries.
upward linking (EM-UPWARD-LINKING)	Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.
verification check (EM-VERIFICATION-CHECK)	An inferred or authored verification artifact that validates requirement behavior with test evidence.
webhook signature (DATA-WEBHOOK-SIGNATURE)	Signature used to verify webhook authenticity.
webhook signature is valid (COND-WEBHOOK-SIGNATURE-VALID)	Condition indicating webhook HMAC verification succeeded. Aliases:signature valid

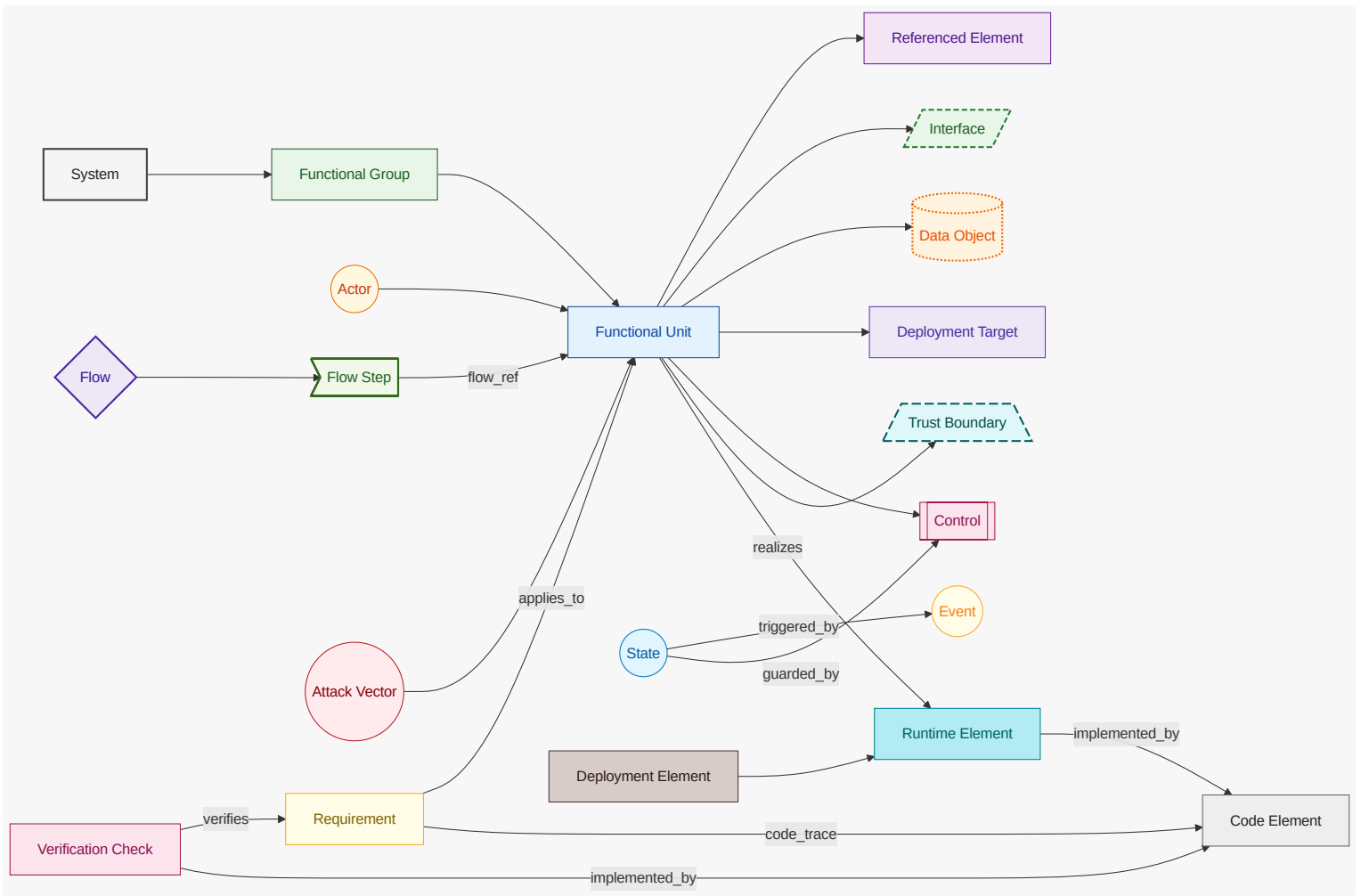
5. Architecture Decision Records

Architecture decision records are generated separately and linked here as a compact index.

ID	Title	Status	Date
ADR-PRR-001	Separate webhook ingress from review orchestration	accepted	2026-04-29

5.1. Diagram Color and Shape Legend

This legend applies to all Mermaid diagrams in this document.



6. Architecture Intent View

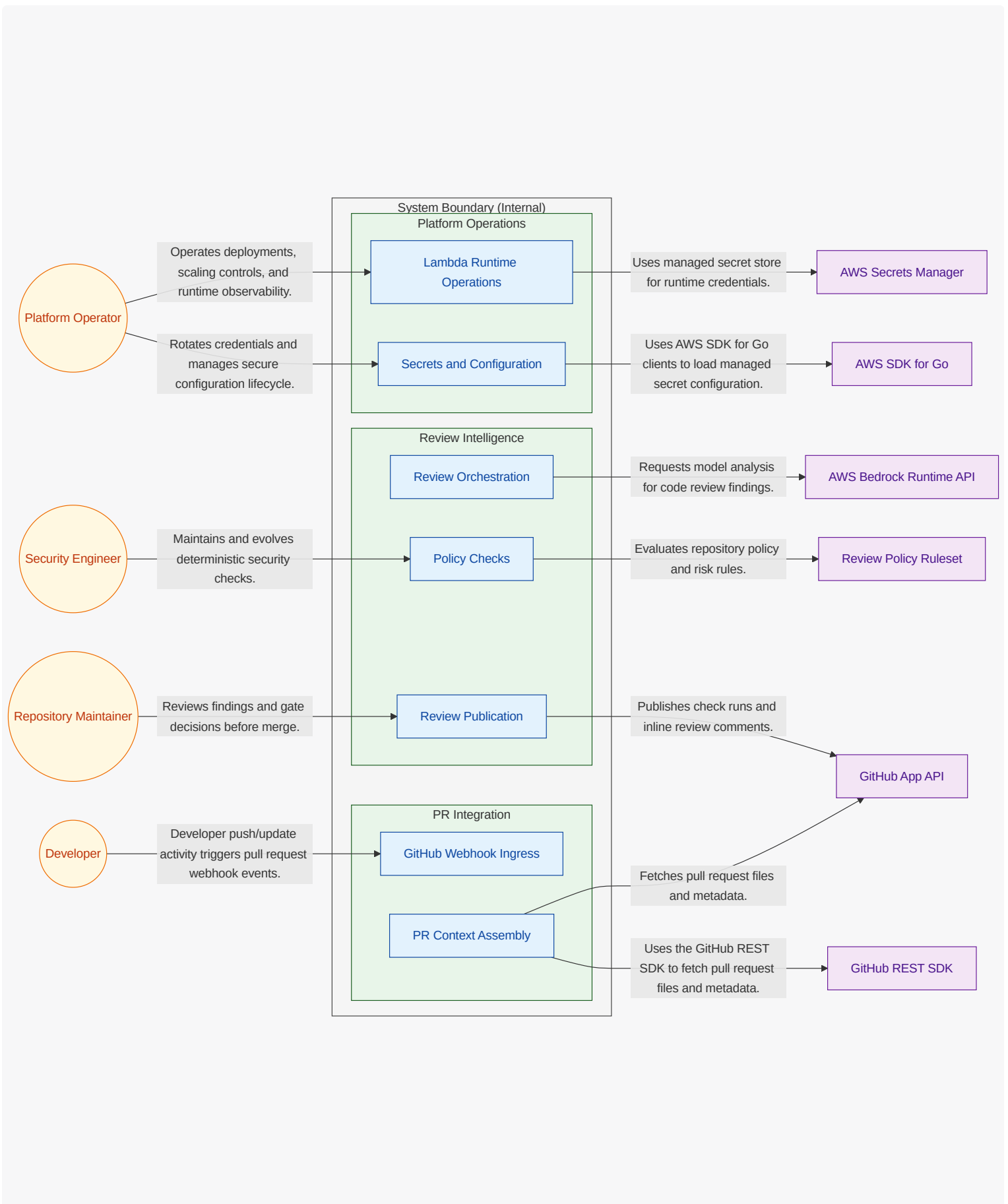
Show the stable authored architecture intent: capability areas, unit ownership boundaries, and intentional responsibility split.

6.1. What This View Answers

- What are the main capability areas?
- What are the stable units of responsibility and what does each unit own?
- How are responsibilities intentionally split?

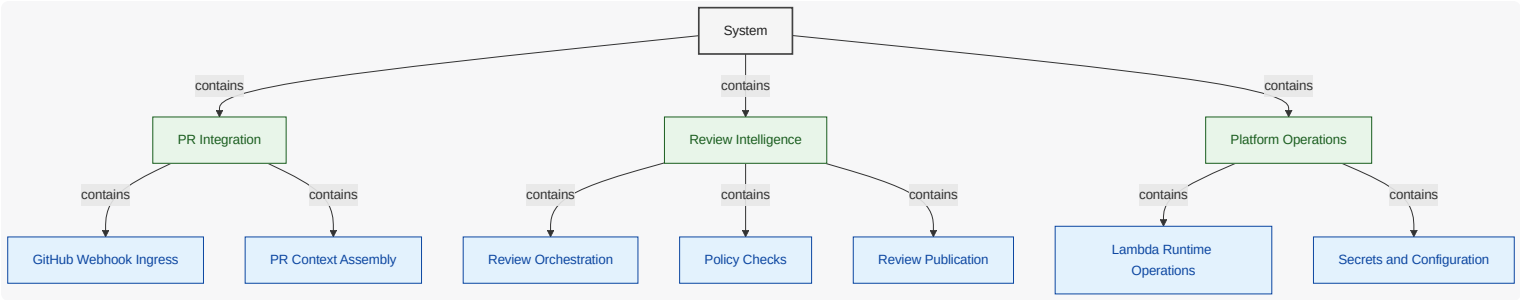
6.2. System Boundary Diagram

What this diagram shows: explicit internal vs external scope. Functional units are inside the **System Boundary (Internal)** container, while actors and referenced dependencies are shown outside the boundary.



6.3. System Decomposition Diagram

What this diagram shows: stable capability decomposition from functional groups into functional units.



6.4. Functional Group Manhattan Matrix

What this diagram shows: a table-like matrix where functional groups are the bottom row (columns) and functional units are arranged in rows above their owning group.

	Policy Checks	
GitHub Webhook Ingress	Review Orchestration	Lambda Runtime Operations
PR Context Assembly	Review Publication	Secrets and Configuration
PR Integration	Review Intelligence	Platform Operations

6.5. Functional Groups and Units (View Scoped)

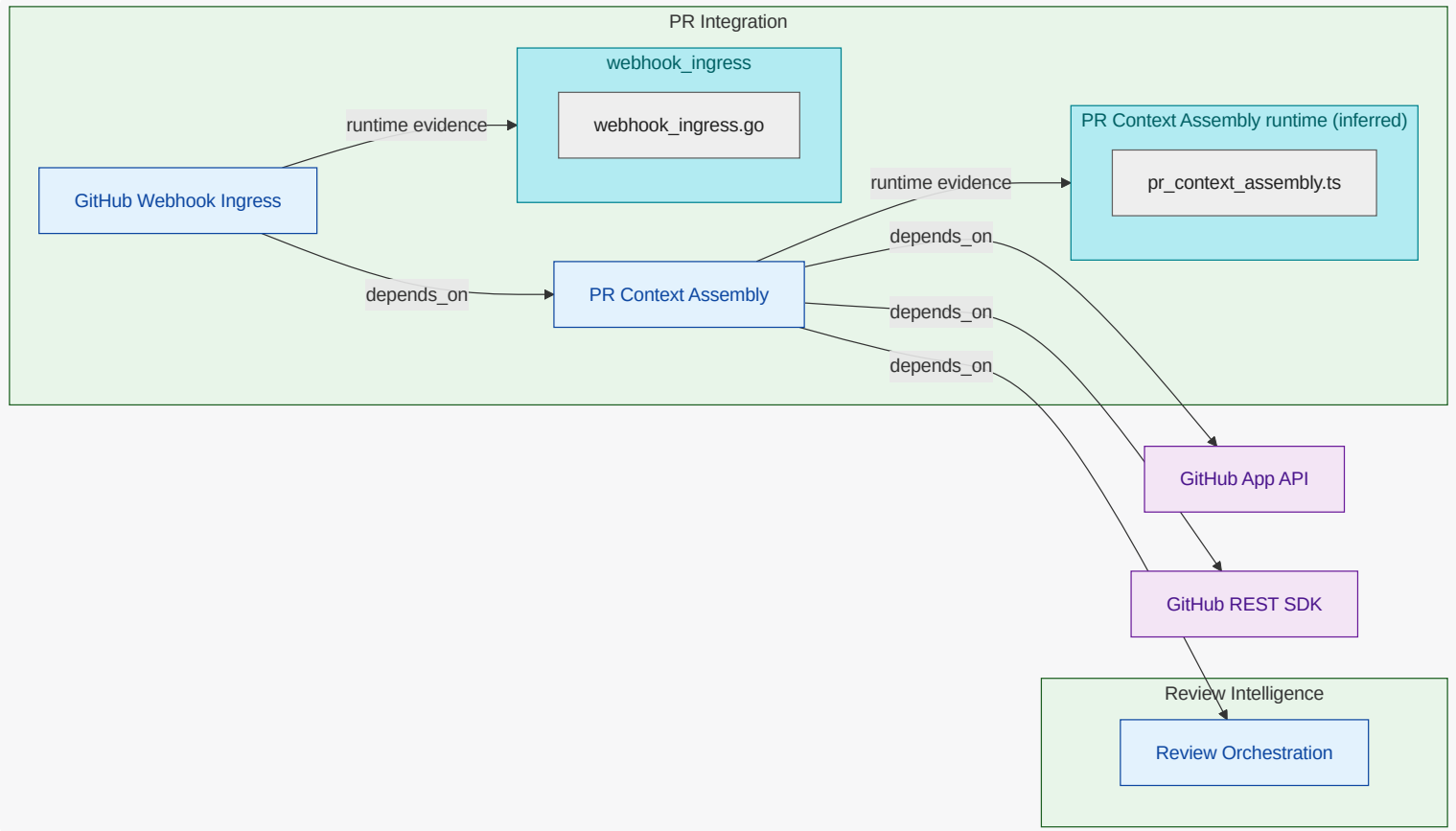
6.5.1. PR Integration (FG-PR-INTEGRATION)

Ingress and context assembly for GitHub pull request events .

<<REF_IDX-CATALOG_FG_PR_INTEGRATION,PR Integration>> owns webhook <<REF_IDX-ENGMODEL_EM_TRUST_BOUNDARY,trust boundaries>> and <<REF_IDX-DO_DO_BEDROCK_PR_CONTEXT,pull request context>> intake. It converts incoming GitHub <<REF_IDX-ENGMODEL_EM_EVENT,events>> into normalized review work with clear boundaries.

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



GitHub Webhook Ingress (FU-GITHUB-WEBHOOK-INGRESS)

Verifies webhook signatures , validates event type, and starts review workflow routing.

Owned Responsibility

functional responsibility for github webhook ingress ; includes decision and orchestration flow to PR Context Assembly

Key Collaborators

PR Context Assembly

Linked Requirements

REQ-PRR-001 , REQ-PRR-008 , REQ-PRR-010

PR Context Assembly (FU-PR-CONTEXT-ASSEMBLY)

Fetches changed files and diff metadata and prepares bounded review context payloads.

Owned Responsibility

functional responsibility for pr context assembly ; includes decision and orchestration flow to GitHub App API and decision and orchestration flow to GitHub REST SDK

Key Collaborators

GitHub App API , GitHub REST SDK , Review Orchestration

Linked Requirements

REQ-PRR-002

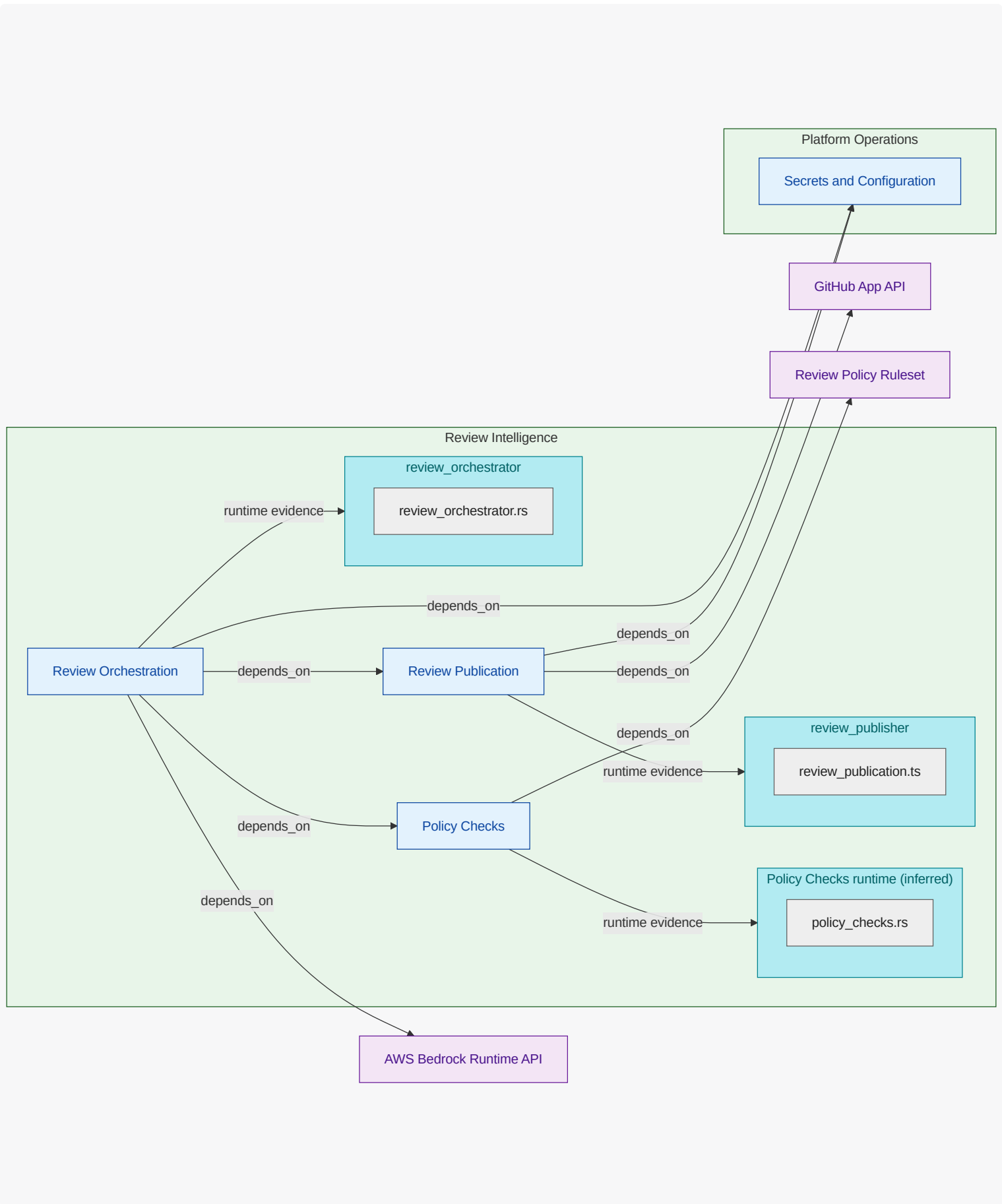
6.5.2. Review Intelligence (FG-REVIEW-INTELLIGENCE)

AI and deterministic review analysis with result publication .

<<REF_IDX-CATALOG_FG_REVIEW_INTELLIGENCE,Review Intelligence>> owns generated findings, <<REF_IDX-CATALOG_DETERMINISTIC_CHECKS,deterministic checks>>, and final <<REF_IDX-CATALOG_PUBLICATION,publication>> behavior. It combines Bedrock reasoning with predictable <<REF_IDX-CATALOG_FU_POLICY_CHECKS,policy checks>> and emits auditable review outputs.

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



Runs **deterministic checks** for secrets, risky IaC changes, and repository policy constraints.

Owned Responsibility

functional responsibility for **policy checks** ; includes decision and **orchestration flow** to **Review Policy Ruleset**

Key Collaborators

Review Policy Ruleset

Linked Requirements

REQ-PRR-004 , **REQ-PRR-006**

Review Orchestration (FU-REVIEW-ORCHESTRATION)

Calls Bedrock models, merges AI and deterministic findings, and handles graceful degradation paths.

Owned Responsibility

functional responsibility for **review orchestration** ; includes decision and **orchestration flow** to **AWS Bedrock Runtime API** and decision and **orchestration flow** to **Policy Checks**

Key Collaborators

AWS Bedrock Runtime API , **Policy Checks** , **Review Publication** , **Secrets and Configuration**

Linked Requirements

REQ-PRR-003 , **REQ-PRR-006**

Review Publication (FU-REVIEW-PUBLICATION)

Publishes check-run summaries and inline comments back to GitHub pull requests.

Owned Responsibility

functional responsibility for **review publication** ; includes decision and **orchestration flow** to **GitHub App API** and decision and **orchestration flow** to **Secrets and Configuration**

Key Collaborators

GitHub App API , **Secrets and Configuration**

Linked Requirements

REQ-PRR-005 , **REQ-PRR-006** , **REQ-PRR-007** , **REQ-PRR-008** , **REQ-PRR-009** , **REQ-PRR-010**

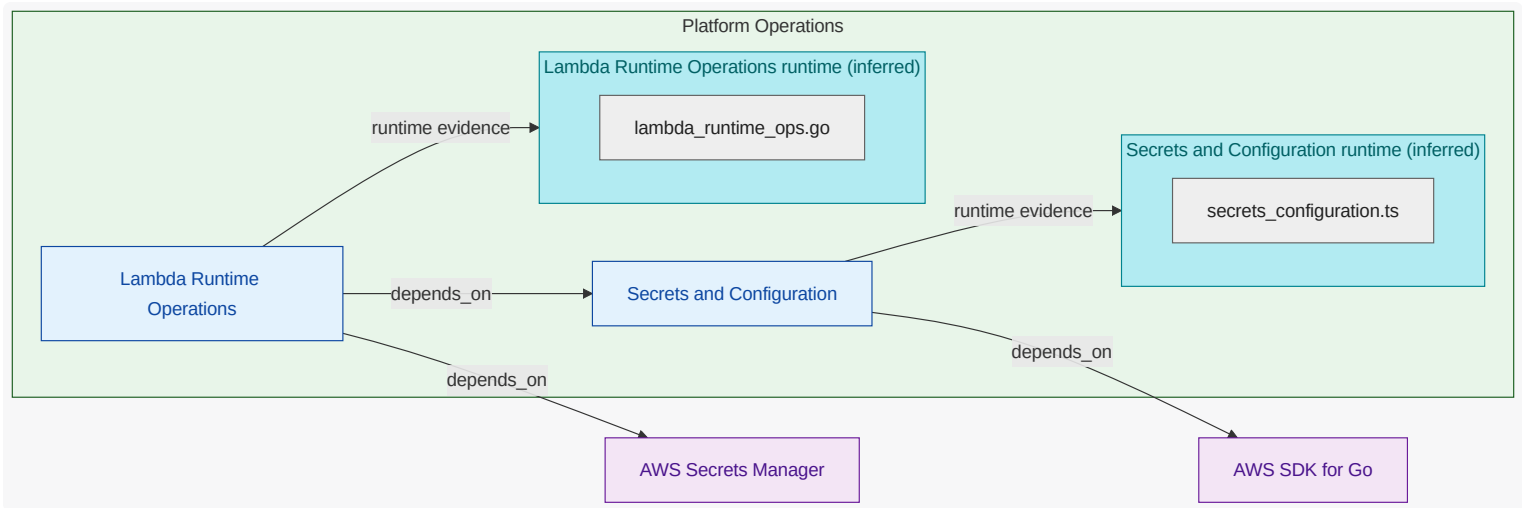
6.5.3. Platform Operations (FG-PLATFORM-OPERATIONS)

Lambda deployment lifecycle, **runtime operations** , and secrets/configuration.

<<REF_IDX-CATALOG_FG_PLATFORM_OPERATIONS,Platform Operations>> owns safe deployment, runtime controls, and secret/config lifecycle for the app.
It enables dependable review operation without owning business-level review policy semantics.

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



Lambda Runtime Operations (FU-LAMBDA-RUNTIME-OPERATIONS)

Manages Lambda packaging/deployment lifecycle and operational health guardrails.

Owned Responsibility

functional responsibility for lambda runtime operations ; includes decision and orchestration flow to AWS Secrets Manager and decision and orchestration flow to Secrets and Configuration

Key Collaborators

AWS Secrets Manager , Secrets and Configuration

Linked Requirements

no explicit authored requirement trigger

Secrets and Configuration (FU-SECRETS-CONFIGURATION)

Manages GitHub App credentials, Bedrock config, and redaction-safe runtime configuration.

Owned Responsibility

functional responsibility for secrets and configuration ; includes decision and orchestration flow to AWS SDK for Go

Key Collaborators

AWS SDK for Go

Linked Requirements

REQ-PRR-007

7. Communication View

Show who communicates with whom across key request paths, including interface edges and dependency handoffs.

7.1. What This View Answers

- Who talks to whom and over which interfaces?
- Which edges are synchronous, asynchronous, event-driven, or callback-oriented?
- Which communication paths are critical to the main request flow?

7.2. Coverage Gaps

Coverage summary: 3/7 units have direct evidence coverage in this view.

- Lambda Runtime Operations has no direct inferred runtime/deployment evidence yet.
- PR Context Assembly has no direct inferred runtime/deployment evidence yet.
- Policy Checks has no direct inferred runtime/deployment evidence yet.
- Secrets and Configuration has no direct inferred runtime/deployment evidence yet.

7.3. Recommended Next Evidence Additions

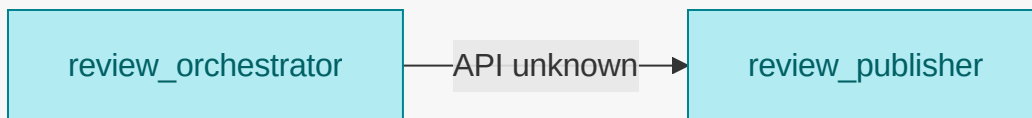
- Add explicit communication metadata (protocol, sync/async/event/callback) to authored mappings where currently implicit.
- Expose service protocol/port metadata in Helm or manifest values to enrich interface-edge inference.
- Capture missing communicating runtime units by adding discoverable workload artifacts under scanned inference roots.

7.4. Communication Path Graph

What this diagram shows: inferred unit-to-unit request/call flow, including available interface/port labels.

NOTE

This graph is inferred from available deployment/runtime artifacts and may be partial. Missing edges or interface details indicate missing derived communication evidence, not necessarily missing behavior.



7.5. Functional Groups and Units (View Scoped)

8. Deployment View

Show inferred deployment artifacts and ownership mapping to authored units. This view emphasizes deployment relationships and platform operations.

8.1. What This View Answers

- How does authored intent become deployed runtime?
- What control-plane objects manage deployable units?
- Where are platform ownership boundaries and missing ownership signals?

8.2. Coverage Gaps

Coverage summary: 3/7 units have direct evidence coverage in this view.

- Lambda Runtime Operations has no direct inferred runtime/deployment evidence yet.
- PR Context Assembly has no direct inferred runtime/deployment evidence yet.
- Policy Checks has no direct inferred runtime/deployment evidence yet.
- Secrets and Configuration has no direct inferred runtime/deployment evidence yet.

8.3. Recommended Next Evidence Additions

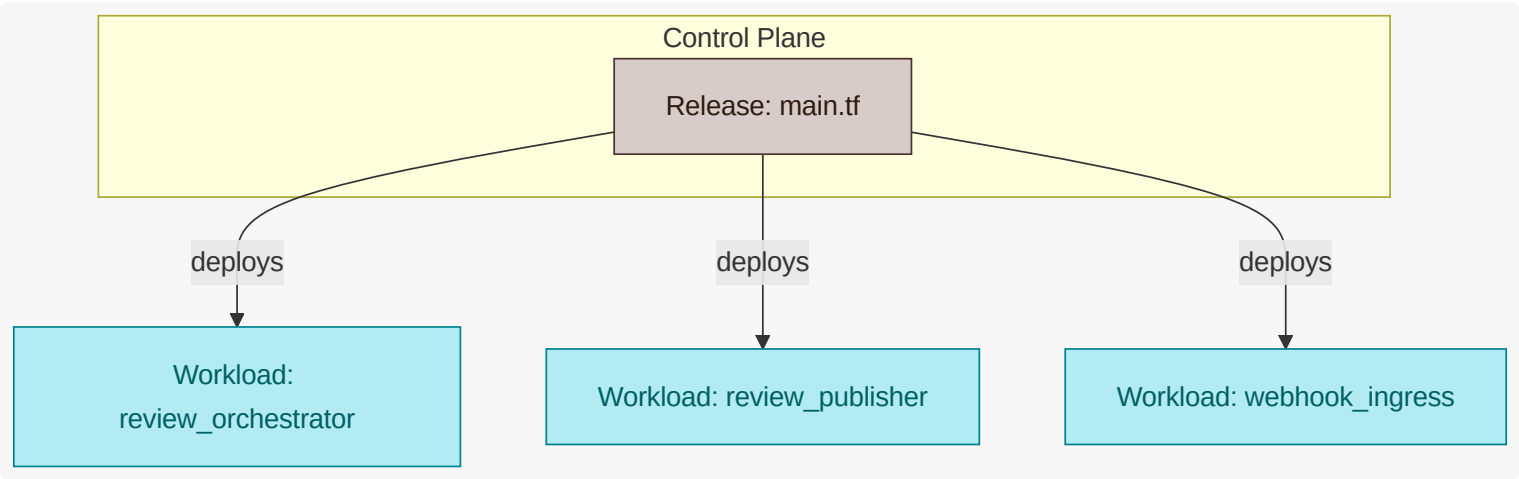
- Add owner metadata to Flux/Kustomization/Helm objects for clearer platform ownership mapping.
- Ensure deployment control artifacts (source, kustomization, release) are all included in scanned directories.
- Add deterministic naming conventions for release and namespace objects to improve cross-artifact linking.

8.4. Deployment Artifact Relationship Graph

What this diagram shows: how source/control-plane artifacts connect to releases, namespaces, workloads, and target cluster context.

NOTE

This graph is inferred from available deployment artifacts (for example Terraform, Flux, Helm) and may be partial when source metadata is incomplete. Namespace and cluster boundaries are rendered as containers when target/placement relationships are inferred. Node labels include role hints (for example Source , Kustomization , Release , Workload , ns/...) to distinguish similarly named elements.

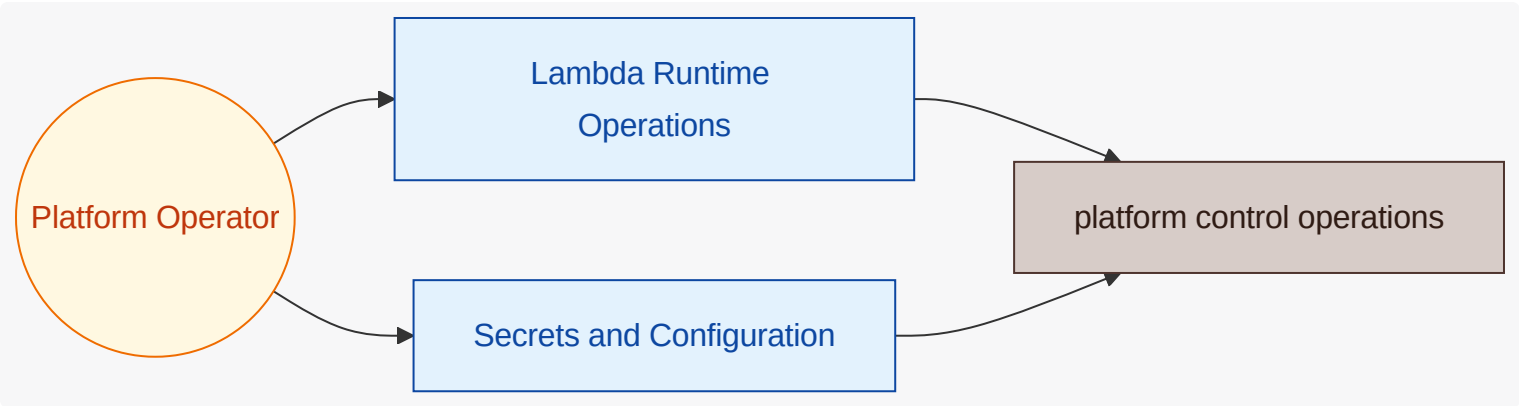


8.5. Platform Operations Graph

What this diagram shows: which platform actors operate which platform units and the control-plane targets they influence.

NOTE

Platform operations links are inferred from authored actor mappings plus discovered deployment control-plane artifacts.



8.6. Functional Groups and Units (View Scoped)

8.6.1. PR Integration (FG-PR-INTEGRATION)

PR Integration functions deploy independently to isolate webhook ingress from context-fetch implementation changes.

GitHub Webhook Ingress (FU-GITHUB-WEBHOOK-INGRESS)

GitHub Webhook Ingress Deployment

Deploys as an independently managed Lambda runtime boundary.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	REQ-PRR-001 , REQ-PRR-008 , REQ-PRR-010
Deployment dependencies	PR Context Assembly

Deployment Aspect	View-Scoped Detail
Inferred deployment evidence	runtime: webhook_ingress

PR Context Assembly (FU-PR-CONTEXT-ASSEMBLY)

PR Context Assembly Deployment

Deployed separately to allow context shaping changes without ingress disruption.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	REQ-PRR-002
Deployment dependencies	GitHub App API , GitHub REST SDK , Review Orchestration
Inferred deployment evidence	code modules: pr_context_assembly

8.6.2. Review Intelligence (FG-REVIEW-INTELLIGENCE)

Review Intelligence units roll independently to preserve availability and reduce blast radius during model or rule updates.

Policy Checks (FU-POLICY-CHECKS)

Policy Checks Deployment

Deployed independently to evolve policy logic without model coupling.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	REQ-PRR-004 , REQ-PRR-006
Deployment dependencies	Review Policy Ruleset
Inferred deployment evidence	code modules: policy_checks

Review Orchestration (FU-REVIEW-ORCHESTRATION)

Review Orchestration Deployment

Isolated deployment enables cautious model/config rollout and fallback control .

Deployment Aspect	View-Scoped Detail
Deployment trigger context	REQ-PRR-003 , REQ-PRR-006
Deployment dependencies	AWS Bedrock Runtime API, Policy Checks , Review Publication , Secrets and Configuration
Inferred deployment evidence	runtime: review_orchestrator

Review Publication (FU-REVIEW-PUBLICATION)

Review Publication Deployment

Release isolation protects publication reliability during other unit rollouts.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	REQ-PRR-005 , REQ-PRR-006 , REQ-PRR-007 , REQ-PRR-008 , REQ-PRR-009 , REQ-PRR-010
Deployment dependencies	GitHub App API , Secrets and Configuration
Inferred deployment evidence	runtime: review_publisher

8.6.3. Platform Operations (FG-PLATFORM-OPERATIONS)

Platform Operations governs release orchestration and runtime placement to keep review capacity stable.

Lambda Runtime Operations (FU-LAMBDA-RUNTIME-OPERATIONS)

Lambda Runtime Operations Deployment

Governs source-to-release pipeline behavior and runtime placement boundaries.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	no explicit authored requirement trigger
Deployment dependencies	AWS Secrets Manager , Secrets and Configuration
Inferred deployment evidence	code modules: lambda_runtime_ops

Secrets and Configuration (FU-SECRETS-CONFIGURATION)

Secrets and Configuration Deployment

Managed as a shared operational boundary with controlled rotation and rollout flow .

Deployment Aspect	View-Scoped Detail
Deployment trigger context	REQ-PRR-007
Deployment dependencies	AWS SDK for Go
Inferred deployment evidence	code modules: secrets_configuration

9. Security View

Show inferred exposure and dependency risk points aligned to unit boundaries, focused on attack paths and security evidence.

9.1. What This View Answers

- What are the main trust boundaries and attack paths?
- Which units are exposed and what security evidence exists?
- Where is the threat model incomplete?

9.2. Coverage Gaps

Coverage summary: 5/7 units have direct evidence coverage in this view.

- Lambda Runtime Operations has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- Policy Checks has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.

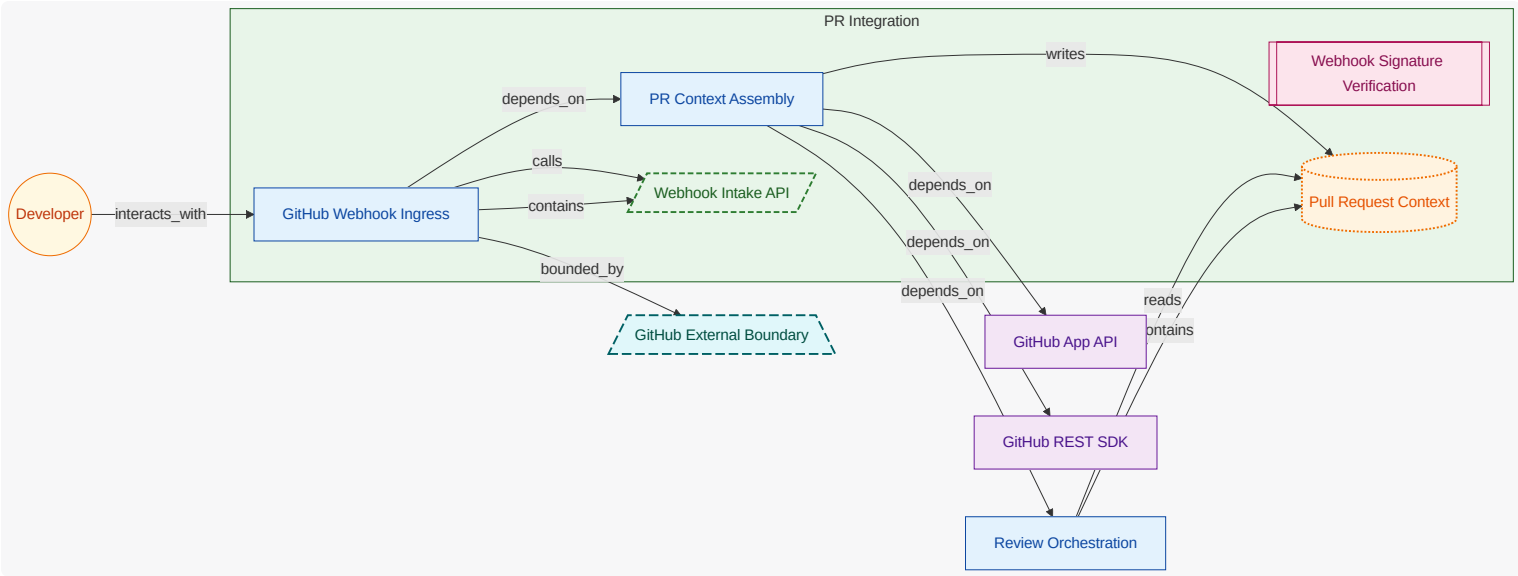
9.3. Recommended Next Evidence Additions

- Add explicit authored attack-vector mappings for units currently marked without threat linkage.
- Add security signal ownership metadata (logs/alerts/ events) for unresolved observability evidence.
- Map additional security-relevant dependencies and exposures into authored mappings for traceable coverage.

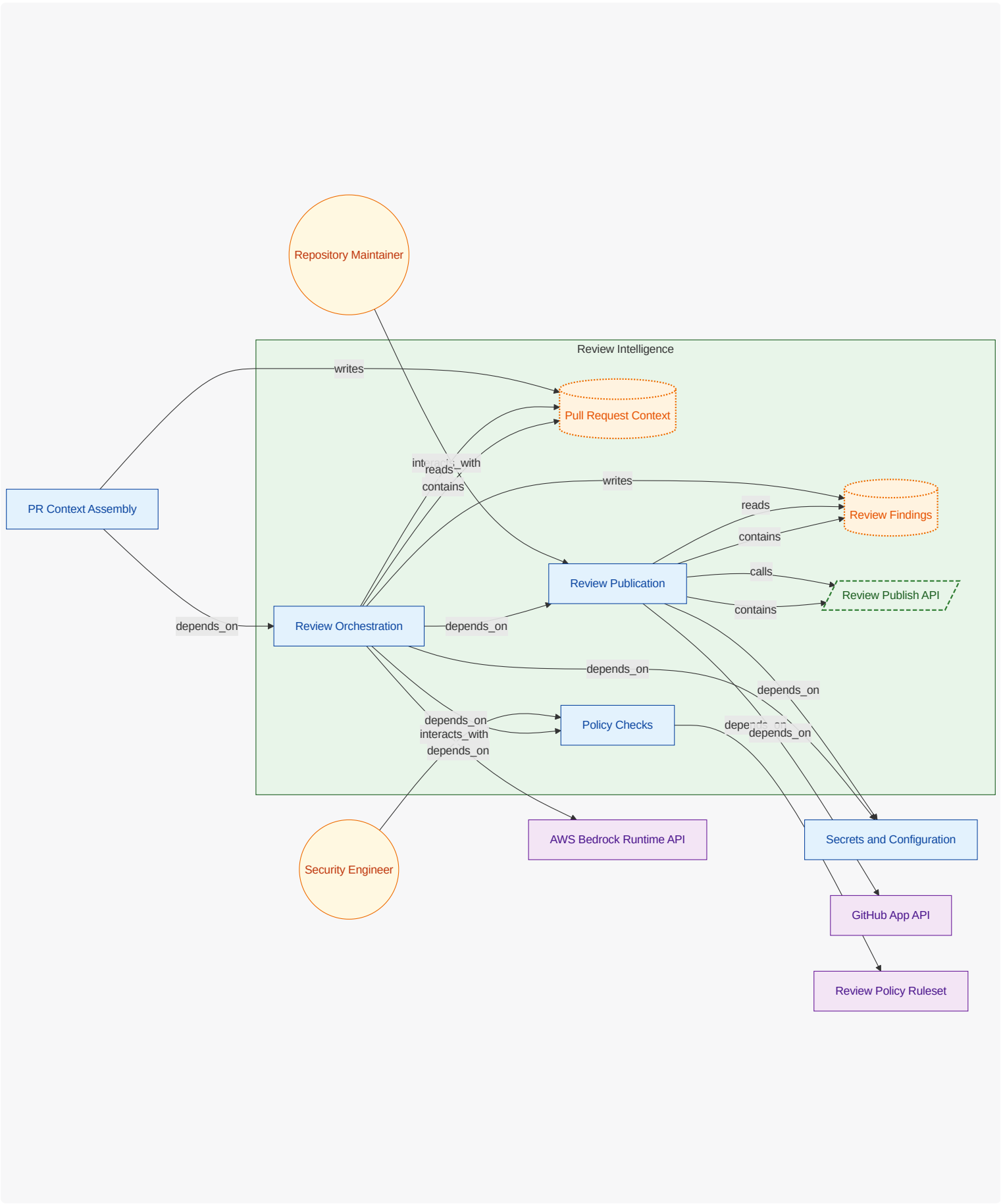
9.4. Threat Model Context Diagram

What these diagrams show: OWASP-style context views split by functional group. Each group diagram keeps owned functional units and owned security-relevant entities inside the functional-group container; connected actors, referenced elements, external functional units, and other external entities are rendered outside it.

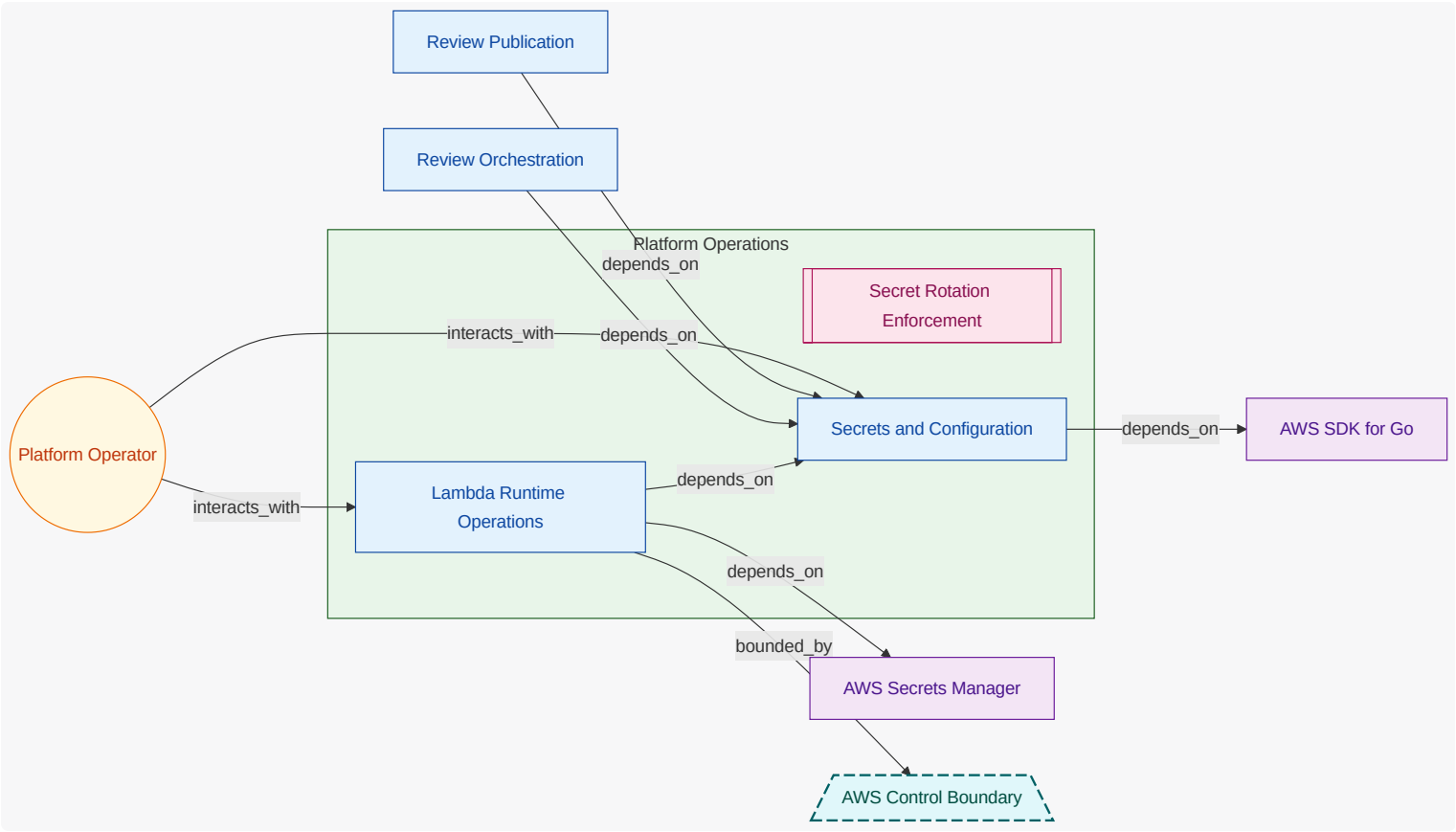
9.4.1. PR Integration (FG-PR-INTEGRATION)



9.4.2. Review Intelligence (FG-REVIEW-INTELLIGENCE)



9.4.3. Platform Operations (FG-PLATFORM-OPERATIONS)



9.5. Threat Scenario Register

9.5.1. Prompt injection in diff influences review output integrity (TS-BEDROCK-PROMPT-INJECTION)

Malicious prompts in diff content attempt to coerce model output away from policy intent.

Field	Value
Status	mitigating
Owner	Security Engineer
Attack Vector	Prompt Injection in Diff
Scope	PR Context Assembly , Pull Request Context , Review Orchestration
Flows	PR Review Interaction Flow
Likelihood	medium
Impact	medium
Severity	medium
Risk	RISK-BEDROCK-PROMPT-INJECTION

Field	Value
Controls	Webhook Signature Verification
Mitigations	MIT-BEDROCK-CONTEXT-SANITIZATION
Verifications	CV-BEDROCK-PROMPT-INJECTION-RESILIENCE
Operational Notes	Track closure through mitigation and verification linkage.

9.5.2. Sensitive tokens leak via review publication payload (TS-BEDROCK-SECRET-LEAK-IN-PUBLICATION)

Review output includes sensitive content due to incomplete redaction before publication .

Field	Value
Status	mitigating
Owner	Security Engineer
Attack Vector	Secret Leakage in Review Output
Scope	Review Findings , Review Publication , Secrets and Configuration
Flows	PR Review Interaction Flow , Security Policy Tuning Flow
Likelihood	low
Impact	high
Severity	medium
Risk	RISK-BEDROCK-SECRET-LEAKAGE
Controls	Secret Rotation Enforcement
Mitigations	MIT-BEDROCK-OUTPUT-REDACTION
Verifications	CV-BEDROCK-REDACTION-CHECK
Operational Notes	Track closure through mitigation and verification linkage.

9.5.3. Spoofed webhook bypasses ingress verification (TS-BEDROCK-WEBHOOK-SPOOF)

Forged pull request webhook attempts to trigger unauthorized review execution.

Field	Value
Status	mitigating
Owner	Security Engineer

Field	Value
Attack Vector	Spoofed GitHub Webhook
Scope	GitHub Webhook Ingress , Webhook Intake API
Flows	PR Review Interaction Flow
Likelihood	medium
Impact	high
Severity	high
Risk	RISK-BEDROCK-WEBHOOK-SPOOFING
Controls	Webhook Signature Verification
Mitigations	MIT-BEDROCK-WEBHOOK-HMAC
Verifications	CV-BEDROCK-WEBHOOK-SIGNATURE
Operational Notes	Track closure through mitigation and verification linkage.

9.6. Threat Mitigation Coverage

ID	Threat Scenario	Control	Status	Effectiveness	Owner	Verification Links	Notes
MIT-BEDROCK-CONTEXT-SANITIZATION	TS-BEDROCK-PROMPT-INJECTION	Webhook Signature Verification	partial	medium	Security Engineer	CV-BEDROCK-PROMPT-INJECTION-RESILIENCE	Context sanitizer and deterministic policy checks reduce injection impact; coverage expansion continues.
MIT-BEDROCK-OUTPUT-REDACTION	TS-BEDROCK-SECRET-LEAK-IN-PUBLICATION	Secret Rotation Enforcement	partial	medium	Security Engineer	CV-BEDROCK-REDACTION-CHECK	Publication redaction pipeline blocks common secret shapes and is being extended.

ID	Threat Scenario	Control	Status	Effectiveness	Owner	Verification Links	Notes
MIT-BEDROCK-WEBHOOK-HMAC	TS-BEDROCK-WEBHOOK-SPOOF	Webhook Signature Verification	verified	high	Security Engineer	CV-BEDROCK-WEBHOOK-SIGNATURE	HMAC signature checks and ingress rejection reduce spoofing attack viability.

9.7. Control Verification Status

ID	Control	Threat Scenarios	Risks	Method	Status	Owner / Last Tested	Findings
CV-BEDROCK-PROMPT-INJECTION-RESILIENCE	Webhook Signature Verification	TS-BEDROCK-PROMPT-INJECTION	RISK-BEDROCK-PROMPT-INJECTION	analysis	partial	Security Engineer / 2026-04-19	deterministic policy merge blocks obvious prompt directives; edge cases remain under review
CV-BEDROCK-REDACTION-CHECK	Secret Rotation Enforcement	TS-BEDROCK-SECRET-LEAK-IN-PUBLICATION	RISK-BEDROCK-SECRET-LEAKAGE	inspection	partial	Security Engineer / 2026-04-21	redaction filters exist; custom token patterns are still being expanded
CV-BEDROCK-WEBHOOK-SIGNATURE	Webhook Signature Verification	TS-BEDROCK-WEBHOOK-SPOOF	RISK-BEDROCK-WEBHOOK-SPOOFING	test	pass	Security Engineer / 2026-04-11	invalid webhook signatures rejected before context assembly

9.8. Security Flow Semantics

ID	Title	Kind	Direction	Frequency	Source	Destination	Protocol
FLOW-BEDROCK-POLICY-TUNING	Security Policy Tuning Flow	control	internal	on-demand	Security Engineer	Review Publication	internal-api

ID	Title	Kind	Direction	Frequency	Source	Destination	Protocol
FLOW-BEDROCK-PR-REVIEW	PR Review Interaction Flow	security	inbound	realtime	Webhook Intake API	Review Publication	https

ID	Auth	Encryption	Integrity	Boundary Crossing	Trust Boundary	Threats	Data
FLOW-BEDROCK-POLICY-TUNING	workforce-sso	tls1.3	signed-policy-bundle	false	none	TS-BEDROCK-SECRET-LEAK-IN-PUBLICATION	Pull Request Context , Review Findings
FLOW-BEDROCK-PR-REVIEW	hmac-signature	tls1.3	signed-payload	true	AWS Control Boundary	TS-BEDROCK-PROMPT-INJECTION, TS-BEDROCK-WEBHOOK-SPOOF	Pull Request Context , Review Findings

9.9. Threat Assumptions

ID	Title	Status	Owner	Applies To	Rationale
ASM-BEDROCK-GITHUB-SIGNATURE-TRUST	GitHub webhook signature secret remains uncompromised	accepted	Security Engineer	AWS Secrets Manager , GitHub Webhook Ingress , Webhook Intake API	Secret rotation and access controls are centrally governed by platform policy.

9.10. Threat Model Out-of-Scope Decisions

ID	Title	Status	Owner	Applies To	Expires On	Reason
OOS-BEDROCK-GITHUB-CLOUD-PLATFORM	GitHub SaaS internal tenant isolation controls	approved	Security Engineer	GitHub App API , GitHub Integration Boundary	2027-12-31	Internal platform isolation and abuse controls for GitHub SaaS are external to this system boundary.

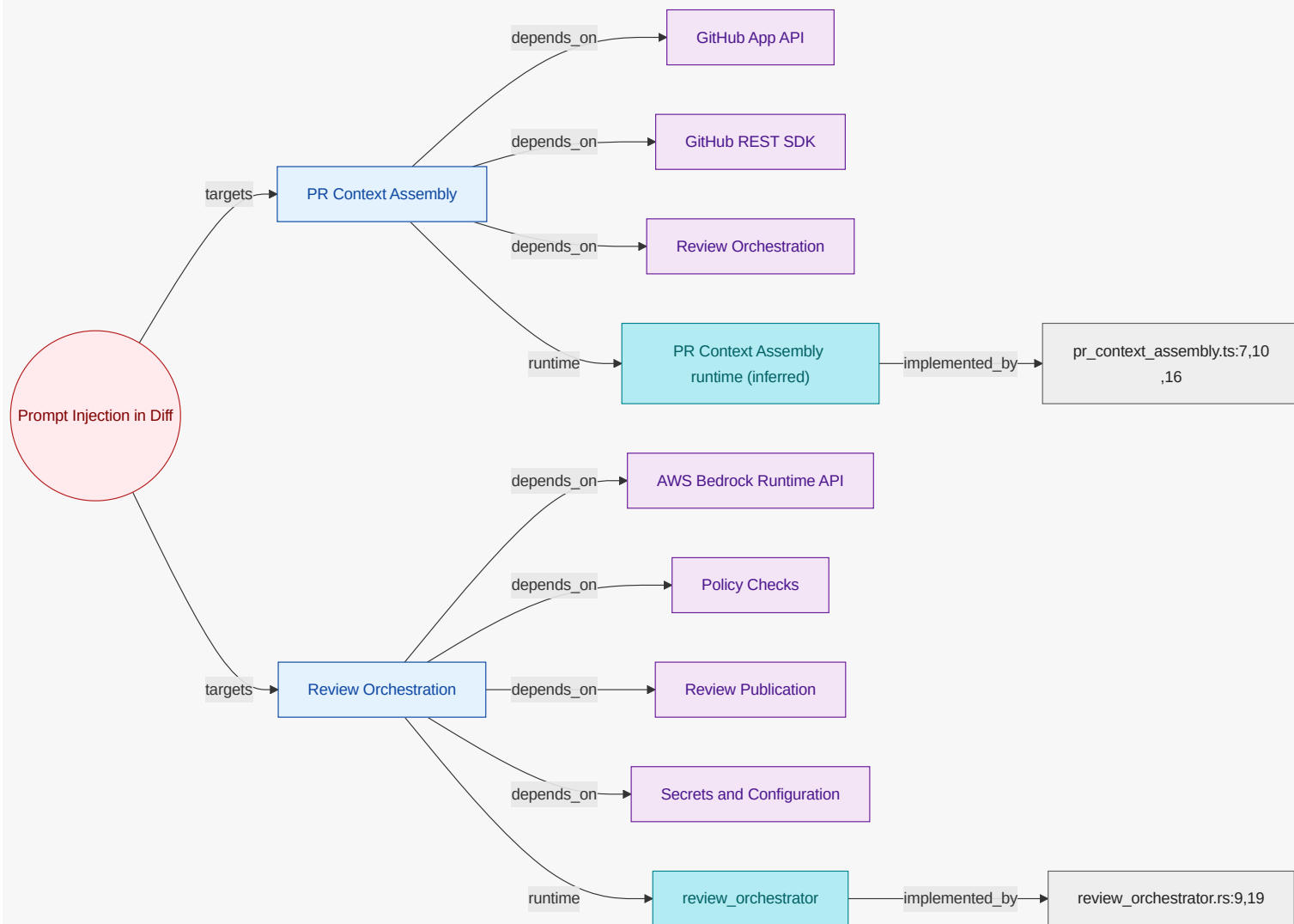
9.11. Attack Vector Chapters

9.11.1. Prompt Injection in Diff (AV-PROMPT-INJECTION-IN-DIFF)

Malicious prompt-like content embedded in pull request diffs.

Mitigated by: none

Trust boundaries: none



PR Context Assembly (FU-PR-CONTEXT-ASSEMBLY)

Applies contextual sanitization to reduce prompt injection risk from diffs.

Attack Surface

GitHub App API , GitHub REST SDK , Review Orchestration

Security Evidence Present

code modules: pr_context_assembly

Review Orchestration (FU-REVIEW-ORCHESTRATION)

Enforces safe prompt construction and bounded output handling under adversarial input.

Attack Surface

AWS Bedrock Runtime API, Policy Checks , Review Publication , Secrets and Configuration

Security Evidence Present

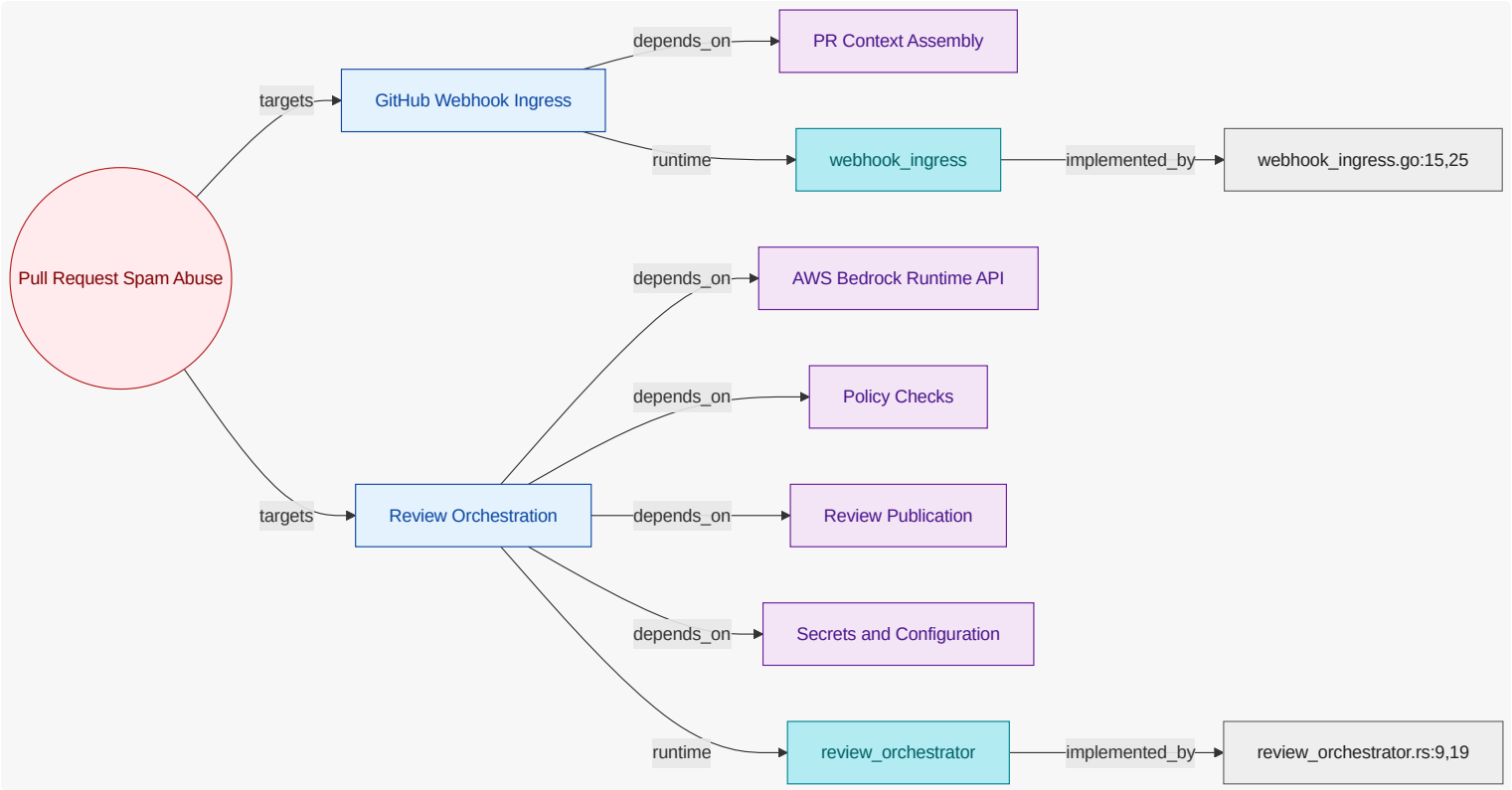
runtime: review_orchestrator | code modules: review_orchestrator

9.11.2. Pull Request Spam Abuse (AV-PR-SPAM-ABUSE)

High-rate PR event flooding that can exhaust review capacity.

Mitigated by: none

Trust boundaries: GitHub External Boundary



GitHub Webhook Ingress (FU-GITHUB-WEBHOOK-INGRESS)

Enforces signature verification and rejects spoofed or malformed events .

Attack Surface

PR Context Assembly

Security Evidence Present

runtime: `webhook_ingress` | code modules: `webhook_ingress`

Review Orchestration (FU-REVIEW-ORCHESTRATION)

Enforces safe prompt construction and bounded output handling under adversarial input.

Attack Surface

AWS Bedrock Runtime API, Policy Checks, Review Publication, Secrets and Configuration

Security Evidence Present

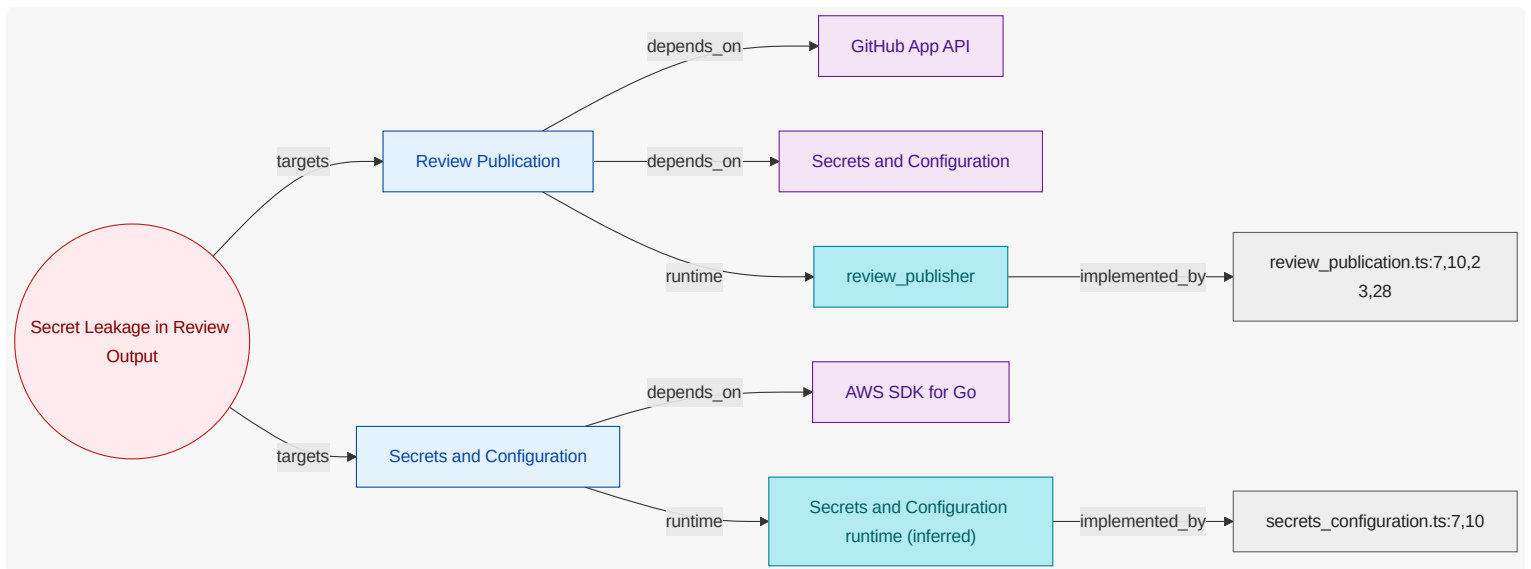
runtime: `review_orchestrator` | code modules: `review_orchestrator`

9.11.3. Secret Leakage in Review Output (AV-SECRET-LEAKAGE-IN-REVIEW)

Generated findings accidentally exposing secrets/tokens in comments.

Mitigated by: Secret Rotation Enforcement

Trust boundaries: none



Review Publication (FU-REVIEW-PUBLICATION)

Applies redaction and policy-aware output filtering before publishing findings.

Attack Surface

GitHub App API, Secrets and Configuration

Security Evidence Present

runtime: `review_publisher` | code modules: `review_publication`

Secrets and Configuration (FU-SECRETS-CONFIGURATION)

Enforces credential isolation, least-privilege access, and redaction-safe metadata handling.

Attack Surface

AWS SDK for Go

Security Evidence Present

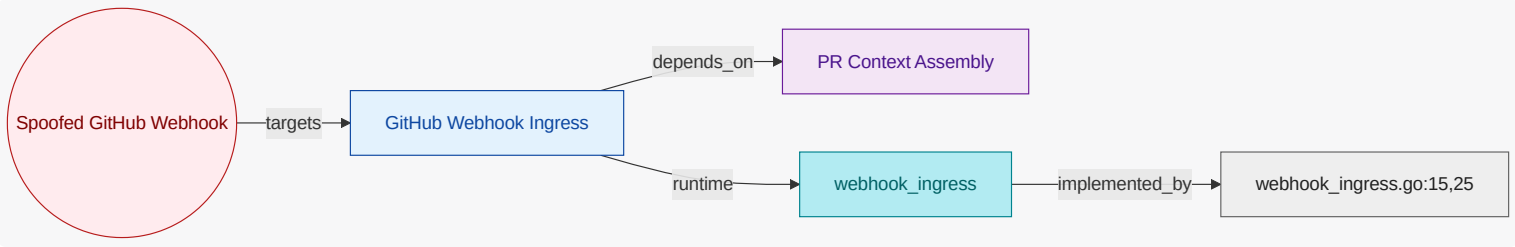
code modules: secrets_configuration

9.11.4. Spoofed GitHub Webhook (AV-SPOOFED-GITHUB-WEBHOOK)

Forged webhook payload attempting unauthorized review execution.

Mitigated by: Webhook Signature Verification

Trust boundaries: GitHub External Boundary



GitHub Webhook Ingress (FU-GITHUB-WEBHOOK-INGRESS)

Enforces signature verification and rejects spoofed or malformed events.

Attack Surface

PR Context Assembly

Security Evidence Present

runtime: webhook_ingress | code modules: webhook_ingress

10. Traceability View

Show requirement-to-unit-to-evidence traceability, including coverage confidence and explicit evidence gaps.

10.1. What This View Answers

- Which `requirements` map to which units?
- Which authored units have runtime/code evidence and which do not?
- Where are confidence or completeness gaps in trace coverage?

10.2. Coverage Gaps

Coverage summary: 7/7 units have direct evidence coverage in this view.

- `Lambda Runtime Operations` has no direct inferred runtime evidence yet.
- `PR Context Assembly` has no direct inferred runtime evidence yet.
- `Policy Checks` has no direct inferred runtime evidence yet.
- `Secrets and Configuration` has no direct inferred runtime evidence yet.

10.3. Recommended Next Evidence Additions

- Add explicit requirement-to-owner mappings where `requirement` scope is currently broad or ambiguous.
- Add ownership annotations for runtime and code artifacts where coverage is still unresolved.
- Prioritize closing gaps for units that are authored but lack both runtime and code evidence.

11. State Lifecycle View

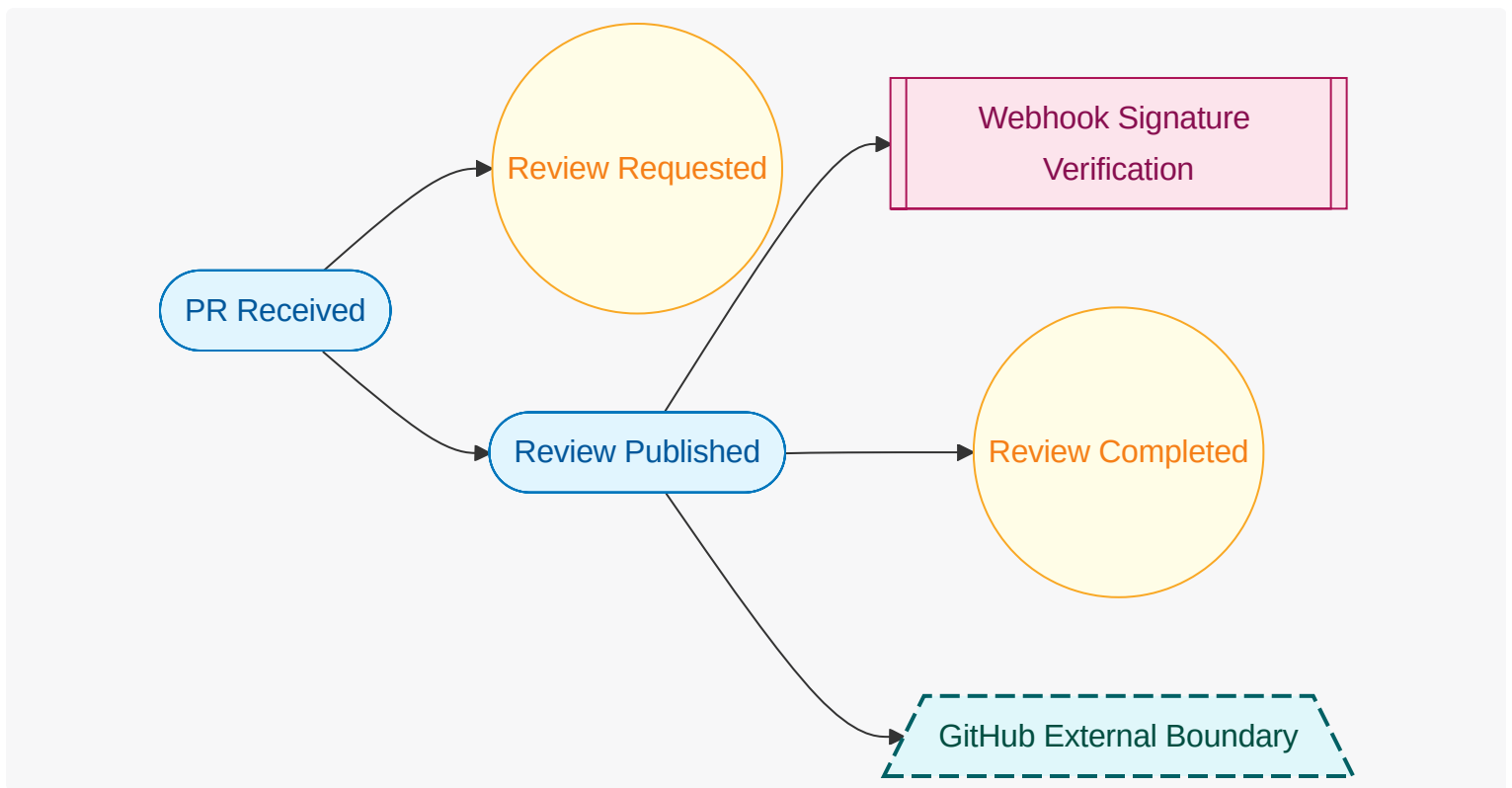
Show workflow state transitions, ownership of transitions, and key exceptional paths.

11.1. What This View Answers

- What states exist and which events trigger transitions?
- Which unit owns each state transition?
- Where do automatic and manual flows diverge?

11.2. State Lifecycle Diagram

What this diagram shows: authored lifecycle states, triggering events, guard controls, trust boundaries, and lifecycle transitions for this view.



11.3. Coverage Gaps

Coverage summary: No functional units are in scope for this view.

- No major coverage gaps detected from current authored and inferred inputs.

11.4. Recommended Next Evidence Additions

- No immediate evidence additions are required for this view.

11.5. Functional Groups and Units (View Scoped)

12. Interaction Flow View

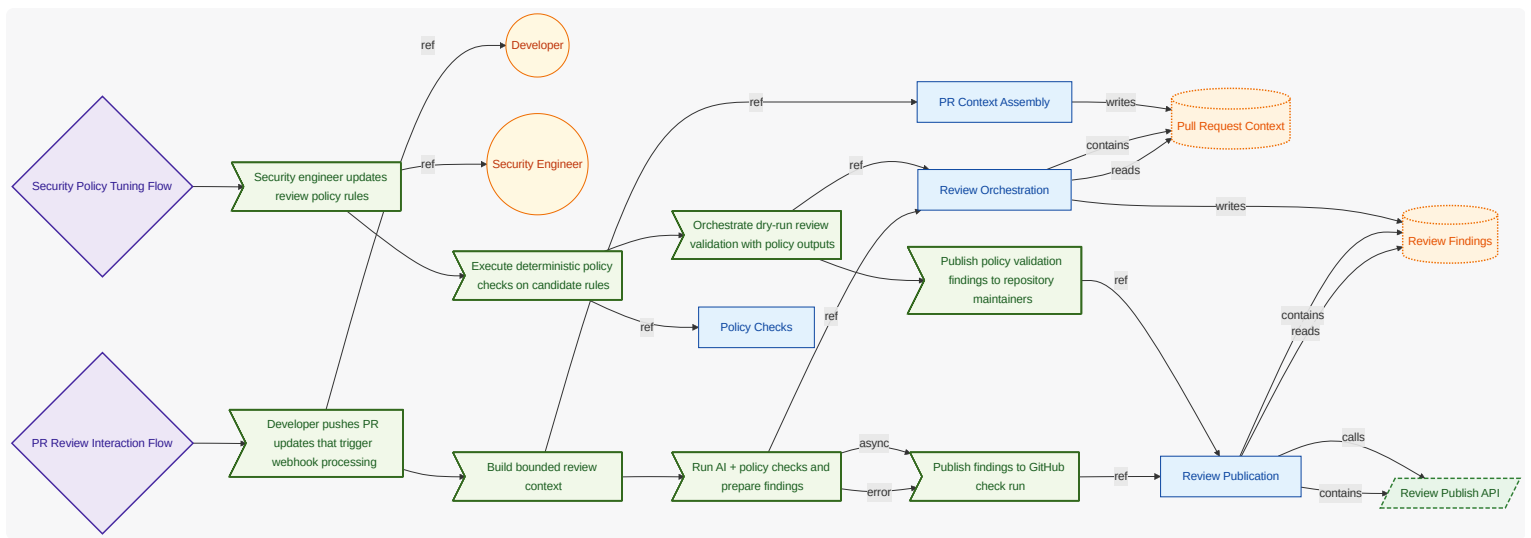
Show user-intent-to-outcome causal flow, including ordered steps, data movement, and error/async branches.

12.1. What This View Answers

- What happens from user intent to system outcome, step by step?
- What data enters/exits each step and where are decisions made?
- Where are async and error paths and what completes each path?

12.2. Interaction Flow Diagram

What this diagram shows: authored interaction flow steps, step ordering, async markers, and referenced actors/interfaces/units for this view.



12.3. Coverage Gaps

Coverage summary: 4/4 units have direct evidence coverage in this view.

- No major coverage gaps detected from current authored and inferred inputs.

12.4. Recommended Next Evidence Additions

- No immediate evidence additions are required for this view.

12.5. Functional Groups and Units (View Scoped)

12.6. Requirement-to-Unit Mapping (Compact)

What this matrix shows: authored requirement-to-unit mappings in a compact square layout (requirements as rows, functional units as columns).

Requirement	FU-GITHUB-WEBHOOK-INGRESS	FU-POLICY-CHECKS	FU-PR-CONTEXT-ASSEMBLY	FU-REVIEW-ORCHESTRATION	FU-REVIEW-PUBLICATION	FU-SECRETS-CONFIGURATION
REQ-PRR-001	X					
REQ-PRR-002			X			
REQ-PRR-003				X		
REQ-PRR-004		X				
REQ-PRR-005					X	
REQ-PRR-006		X		X	X	
REQ-PRR-007					X	X
REQ-PRR-008	X				X	
REQ-PRR-009					X	
REQ-PRR-010	X				X	

12.7. Verification Result Mapping

What this section shows: per-requirement result mapping inferred from test-result artifacts.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-E2E-PR_OPENED_REVIEW_FLOW-6BF0B5	REQ-PRR-001	pass	test-results/e2e-pr-opened-review.xml
VER-INF-E2E-PR_OPENED_REVIEW_FLOW-6BF0B5	REQ-PRR-002	pass	test-results/e2e-pr-opened-review.xml
VER-INF-E2E-PR_OPENED_REVIEW_FLOW-6BF0B5	REQ-PRR-003	pass	test-results/e2e-pr-opened-review.xml
VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-76E20A	REQ-PRR-004	pass	test-results/integration-policy-fallback.json; deterministic policy checks executed
VER-INF-E2E-PR_OPENED_REVIEW_FLOW-6BF0B5	REQ-PRR-005	pass	test-results/contract-checkrun.json; check run payload includes summary and neutral conclusion
VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-17DBA6	REQ-PRR-005	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-76E20A	REQ-PRR-006	pass	test-results/integration-policy-fallback.json; policy-only fallback published
VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-B108A1	REQ-PRR-007	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-DC9EB2	REQ-PRR-008	pass	test-results/unit-rate-limit.xml

13.Traceability Appendix

13.1. Requirement Details

13.1.1. REQ-PRR-001

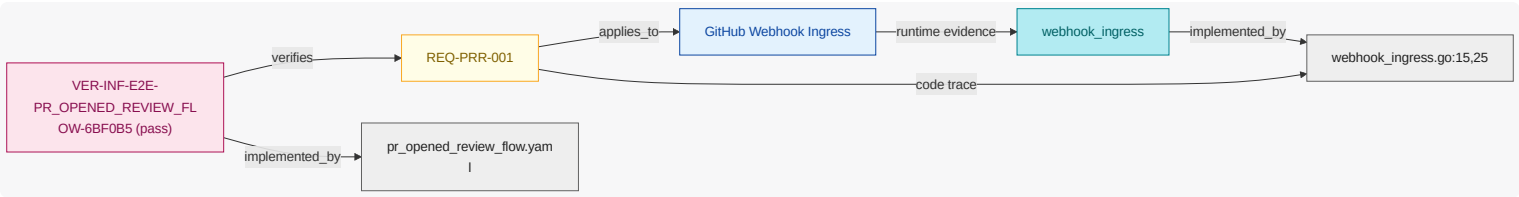
While normal mode is active , when pull request synchronize event is received and webhook signature is valid , the pr review system shall verify webhook signature before processing review workflow.

NOTE

Webhook trust boundary validation.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



13.1.2. REQ-PRR-002

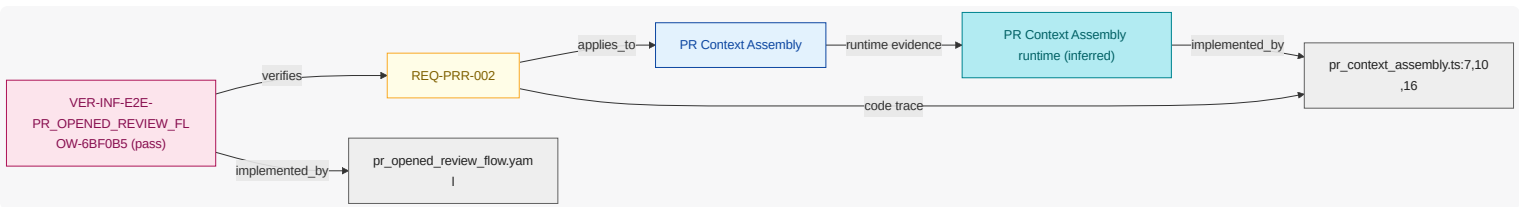
While pull request review is active , when pull request opened event is received , the pr review system shall construct diff context from changed files for developer review execution.

NOTE

Context ingestion from GitHub API .

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



13.1.3. REQ-PRR-003

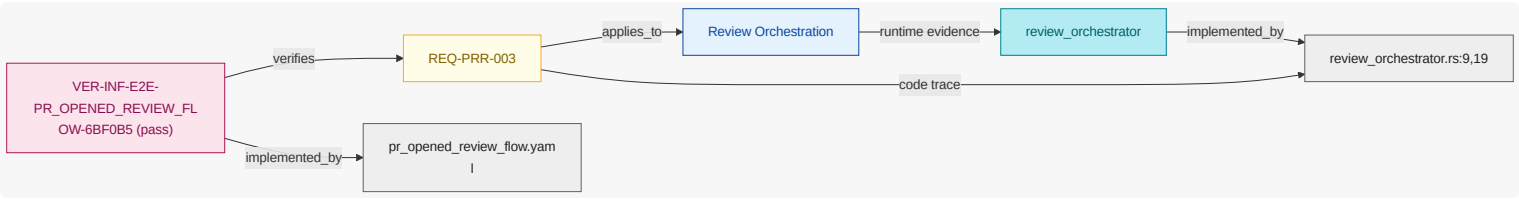
Where ai review is enabled , when diff context is prepared , the pr review system shall request bedrock runtime analysis for review findings .

NOTE

AI-assisted review path.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



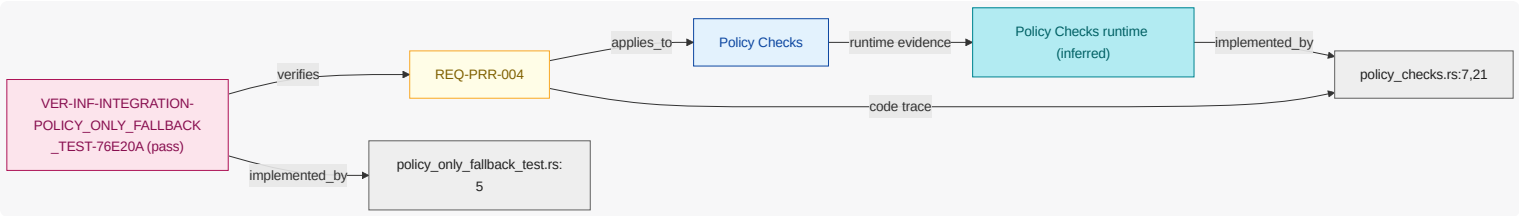
13.1.4. REQ-PRR-004

Where policy checks are enabled , when pull request review is active , the pr review system shall evaluate deterministic policy checks for secret exposure plus infrastructure risk plus dependency changes.

NOTE | Deterministic non-LLM review checks.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



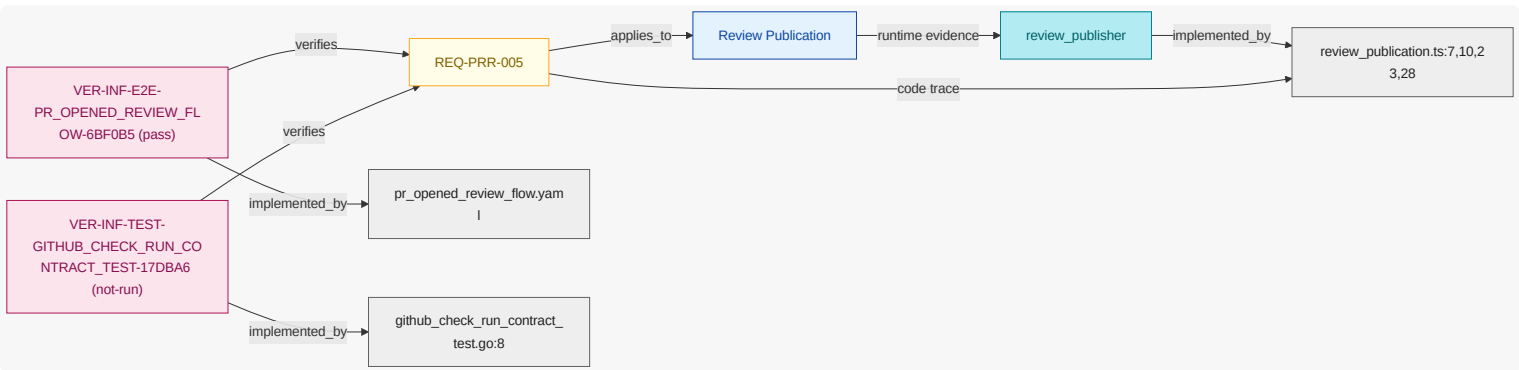
13.1.5. REQ-PRR-005

While pull request review is active , when review findings are generated , the pr review system shall publish check run summary to repository maintainer .

NOTE | Result publication contract.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



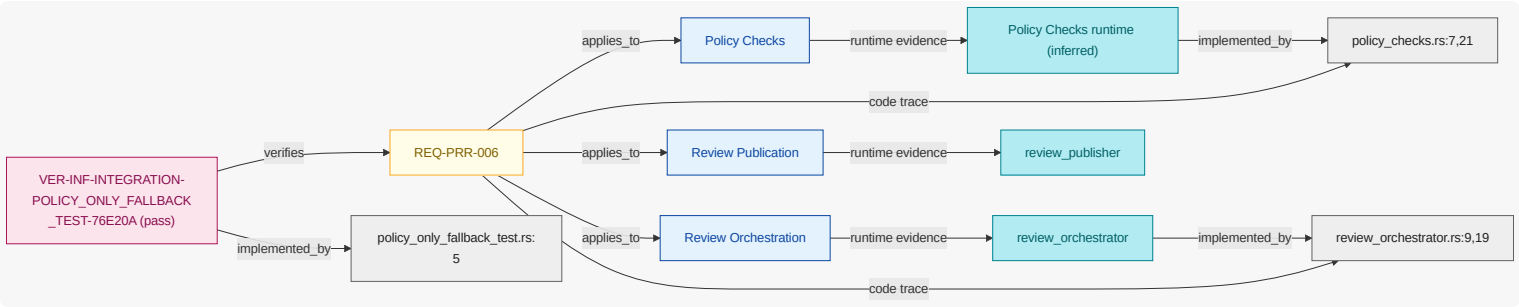
13.1.6. REQ-PRR-006

If bedrock runtime is unavailable , then the pr review system shall publish deterministic policy-only review outcome .

NOTE | Graceful degradation requirement .

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



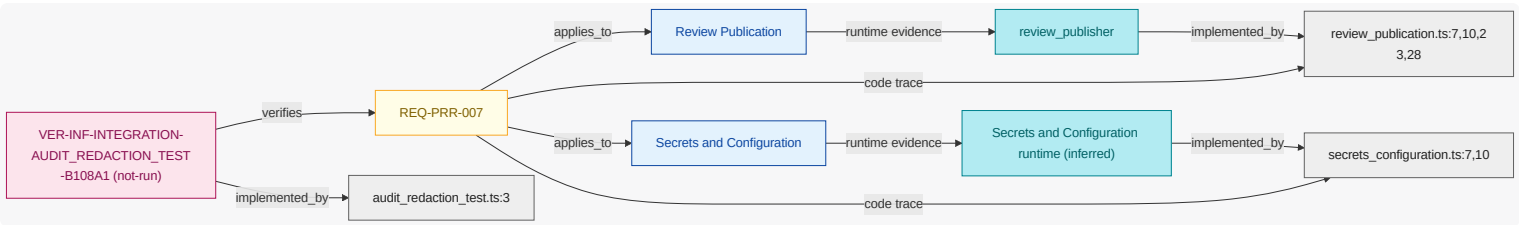
13.1.7. REQ-PRR-007

Where audit logging is enabled, when review findings are published, the pr review system shall persist redacted audit metadata for security engineer audit traceability.

NOTE Evidence and audit lineage.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



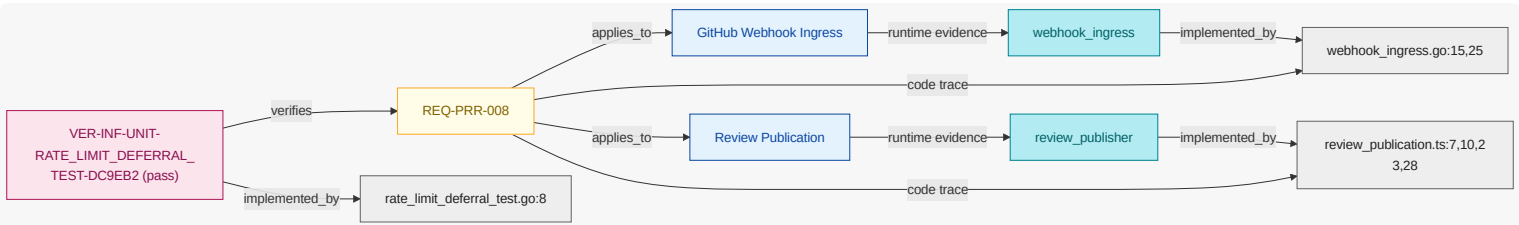
13.1.8. REQ-PRR-008

If review request rate limit is exceeded, then the pr review system shall defer processing.

NOTE Abuse and capacity protection behavior.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



13.1.9. REQ-PRR-009

While pull request review is active , when review findings are generated , the pr review system shall publish inline comments with result metadata to repository maintainer .

NOTE

Inline-annotation publication contract.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



13.1.10. REQ-PRR-010

If review request rate limit is exceeded , then the pr review system shall notify platform operator .

NOTE

Maintainer notification for deferred review processing.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



13.2. Verification Inventory

What this section shows: verification checks inferred from test artifacts, linked to test code elements and mapped to verified requirements.

13.2.1. Pr Opened Review Flow

Field	Value
ID	VER-INF-E2E-PR_OPENED_REVIEW_FLOW-6BF0B5
Kind	e2e
Status	pass
Verifies Requirements	REQ-PRR-001 , REQ-PRR-002 , REQ-PRR-003 , REQ-PRR-005
Test Code Elements	tests/e2e/pr_opened_review_flow.yaml

Field	Value
Derived Functional Owners	FU-GITHUB-WEBHOOK-INGRESS, FU-PR-CONTEXT-ASSEMBLY, FU-REVIEW-ORCHESTRATION, FU-REVIEW-PUBLICATION
Result Summary	pass:4
Evidence	test-results/contract-checkrun.json, test-results/e2e-pr-opened-review.xml, tests/e2e/pr_opened_review_flow.yaml
Notes	exercises end-to-end pull request opened flow from webhook ingest to published review

13.2.2. Audit Redaction Test

Field	Value
ID	VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-B108A1
Kind	integration
Status	not-run
Verifies Requirements	REQ-PRR-007
Test Code Elements	audit_redaction_test.ts:3
Derived Functional Owners	FU-REVIEW-PUBLICATION, FU-SECRETS-CONFIGURATION
Result Summary	not-run:1
Evidence	tests/integration/audit_redaction_test.ts
Notes	checks audit evidence redaction for secrets and sensitive payload fields

13.2.3. Policy Only Fallback Test

Field	Value
ID	VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-76E20A
Kind	integration
Status	pass
Verifies Requirements	REQ-PRR-004, REQ-PRR-006
Test Code Elements	policy_only_fallback_test.rs:5
Derived Functional Owners	FU-POLICY-CHECKS, FU-REVIEW-ORCHESTRATION, FU-REVIEW-PUBLICATION

Field	Value
Result Summary	pass:2
Evidence	test-results/integration-policy-fallback.json, tests/integration/policy_only_fallback_test.rs
Notes	verifies policy-only fallback when Bedrock inference is unavailable

13.2.4. Github Check Run Contract Test

Field	Value
ID	VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-17DBA6
Kind	test
Status	not-run
Verifies Requirements	REQ-PRR-005
Test Code Elements	github_check_run_contract_test.go:8
Derived Functional Owners	FU-REVIEW-PUBLICATION
Result Summary	not-run:1
Evidence	tests/contract/github_check_run_contract_test.go
Notes	validates GitHub check-run contract shape and required review result fields

13.2.5. Rate Limit Deferral Test

Field	Value
ID	VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-DC9EB2
Kind	unit
Status	pass
Verifies Requirements	REQ-PRR-008
Test Code Elements	rate_limit_deferral_test.go:8
Derived Functional Owners	FU-GITHUB-WEBHOOK-INGRESS, FU-REVIEW-PUBLICATION
Result Summary	pass:1
Evidence	test-results/unit-rate-limit.xml, tests/unit/rate_limit_deferral_test.go

Field	Value
Notes	verifies rate-limit deferral behavior for pull request review processing

14. Reference Index

14.1. Authored References

CTRL-BEDROCK-SECRETS-ROTATION

category=credential-management; description=Rotate app and model credentials according to policy.

Key	Value
Kind	Control
Name	Secret Rotation Enforcement
Mentioned In	Security View

CTRL-BEDROCK-WEBHOOK-SIGNATURE

category=input-integrity; description=Enforce signature verification for all GitHub webhook events.

Key	Value
Kind	Control
Name	Webhook Signature Verification
Mentioned In	Security View

DO-BEDROCK-PR-CONTEXT

termRef=DATA-PR-CONTEXT; schemaRef=schemas/pr-context.json; sensitivity=internal

Key	Value
Kind	Data Object
Name	Pull Request Context
Mentioned In	Architecture Intent View , Security View

DO-BEDROCK-REVIEW-FINDINGS

termRef=DATA-REVIEW-FINDINGS; schemaRef=schemas/review-findings.json; sensitivity=internal

Key	Value
Kind	Data Object
Name	Review Findings

Key	Value
Mentioned In	Security View , REQ-PRR-003

DEP-BEDROCK-GITHUB-BOUNDARY

environment=external; region=global; account=github; cluster=external; namespace=app; trustZone=external

Key	Value
Kind	Deployment Target
Name	GitHub Integration Boundary
Mentioned In	Security View

DEP-BEDROCK-LAMBDA-PRIMARY

environment=prod; region=us-east-1; account=app; cluster=lambda; namespace=review; trustZone=app

Key	Value
Kind	Deployment Target
Name	Primary Lambda Runtime
Mentioned In	n/a

EVT-BEDROCK-REVIEW-COMPLETED

Trigger indicating publication complete.

Key	Value
Kind	Event
Name	Review Completed
Mentioned In	n/a

EVT-BEDROCK-REVIEW-REQUESTED

Trigger for analysis orchestration.

Key	Value
Kind	Event
Name	Review Requested
Mentioned In	n/a

FLOW-BEDROCK-POLICY-TUNING

entry=update-policy-rules; exits=publish-policy-validation; steps=4

Key	Value
Kind	Flow
Name	Security Policy Tuning Flow
Mentioned In	Security View

FLOW-BEDROCK-PR-REVIEW

entry=receive-webhook; exits=publish-review; steps=4

Key	Value
Kind	Flow
Name	PR Review Interaction Flow
Mentioned In	Security View

FLOW-BEDROCK-POLICY-TUNING::orchestrate-policy-validation

kind=system_action; ref=FU-REVIEW-ORCHESTRATION; action=Orchestrate dry-run review validation with policy outputs; dataIn=policy_check_results; dataOut=validation_findings; async=false

Key	Value
Kind	Flow Step
Name	Orchestrate dry-run review validation with policy outputs
Mentioned In	n/a

FLOW-BEDROCK-POLICY-TUNING::publish-policy-validation

kind=system_action; ref=FU-REVIEW-PUBLICATION; action=Publish policy validation findings to repository maintainers; dataIn=validation_findings; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Publish policy validation findings to repository maintainers
Mentioned In	n/a

FLOW-BEDROCK-POLICY-TUNING::run-policy-checks

kind=system_action; ref=FU-POLICY-CHECKS; action=Execute deterministic policy checks on candidate rules; dataIn=; dataOut=policy_check_results; async=false

Key	Value
Kind	Flow Step
Name	Execute deterministic policy checks on candidate rules
Mentioned In	n/a

FLOW-BEDROCK-POLICY-TUNING::update-policy-rules

kind=user_action; ref=ACT-SECURITY-ENGINEER; action=Security engineer updates review policy rules; dataIn=; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Security engineer updates review policy rules
Mentioned In	n/a

FLOW-BEDROCK-PR-REVIEW::assemble-context

kind=system_action; ref=FU-PR-CONTEXT-ASSEMBLY; action=Build bounded review context; dataIn=; dataOut=pr_context; async=false

Key	Value
Kind	Flow Step
Name	Build bounded review context
Mentioned In	n/a

FLOW-BEDROCK-PR-REVIEW::orchestrate-review

kind=system_action; ref=FU-REVIEW-ORCHESTRATION; action=Run AI + policy checks and prepare findings; dataIn=pr_context; dataOut=review_findings; async=true

Key	Value
Kind	Flow Step
Name	Run AI + policy checks and prepare findings
Mentioned In	n/a

FLOW-BEDROCK-PR-REVIEW::publish-review

kind=system_action; ref=FU-REVIEW-PUBLICATION; action=Publish findings to GitHub check run; dataIn=review_findings; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Publish findings to GitHub check run
Mentioned In	n/a

FLOW-BEDROCK-PR-REVIEW::receive-webhook

kind=user_action; ref=ACT-DEVELOPER; action=Developer pushes PR updates that trigger webhook processing; dataIn=; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Developer pushes PR updates that trigger webhook processing
Mentioned In	n/a

IF-BEDROCK-REVIEW-PUBLISH-API

protocol=https; endpoint=/github/check-runs; schemaRef=schemas/review-output.json; owner=FU-REVIEW-PUBLICATION

Key	Value
Kind	Interface
Name	Review Publish API
Mentioned In	n/a

IF-BEDROCK-WEBHOOK-API

protocol=https; endpoint=/webhook/github; schemaRef=schemas/webhook-event.json; owner=FU-GITHUB-WEBHOOK-INGRESS

Key	Value
Kind	Interface
Name	Webhook Intake API
Mentioned In	Security View

While normal mode is active, when pull request synchronize event is received and webhook signature is valid, the pr review system shall verify webhook signature before processing review workflow.

Key	Value
Kind	Requirement
Name	REQ-PRR-001
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Pr Opened Review Flow

REQ-PRR-002

While pull request review is active, when pull request opened event is received, the pr review system shall construct diff context from changed files for developer review execution.

Key	Value
Kind	Requirement
Name	REQ-PRR-002
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Pr Opened Review Flow

REQ-PRR-003

Where ai review is enabled, when diff context is prepared, the pr review system shall request bedrock runtime analysis for review findings.

Key	Value
Kind	Requirement
Name	REQ-PRR-003
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Pr Opened Review Flow

REQ-PRR-004

Where policy checks are enabled, when pull request review is active, the pr review system shall evaluate deterministic policy checks for secret exposure plus infrastructure risk plus dependency changes.

Key	Value
Kind	Requirement
Name	REQ-PRR-004
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Policy Only Fallback Test

REQ-PRR-005

While pull request review is active, when review findings are generated, the pr review system shall publish check run summary to repository maintainer.

Key	Value
Kind	Requirement
Name	REQ-PRR-005
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Pr Opened Review Flow , Github Check Run Contract Test

REQ-PRR-006

If bedrock runtime is unavailable, then the pr review system shall publish deterministic policy-only review outcome.

Key	Value
Kind	Requirement
Name	REQ-PRR-006
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Policy Only Fallback Test

REQ-PRR-007

Where audit logging is enabled, when review findings are published, the pr review system shall persist redacted audit metadata for security engineer audit traceability.

Key	Value
Kind	Requirement
Name	REQ-PRR-007
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Audit Redaction Test

REQ-PRR-008

If review request rate limit is exceeded, then the pr review system shall defer processing.

Key	Value
Kind	Requirement
Name	REQ-PRR-008
Mentioned In	Architecture Intent View , Deployment View , Interaction Flow View , Rate Limit Deferral Test

REQ-PRR-009

While pull request review is active, when review findings are generated, the pr review system shall publish inline comments with result metadata to repository maintainer.

Key	Value
Kind	Requirement
Name	REQ-PRR-009
Mentioned In	Architecture Intent View , Deployment View

REQ-PRR-010

If review request rate limit is exceeded, then the pr review system shall notify platform operator.

Key	Value
Kind	Requirement
Name	REQ-PRR-010
Mentioned In	Architecture Intent View , Deployment View

STATE-BEDROCK-PR-RECEIVED

PR webhook accepted and queued for analysis.

Key	Value
Kind	State
Name	PR Received
Mentioned In	n/a

STATE-BEDROCK-REVIEW-PUBLISHED

Findings published to GitHub check run output.

Key	Value
Kind	State
Name	Review Published
Mentioned In	n/a

TB-BEDROCK-AWS-CONTROL

Boundary between app workloads and privileged platform controls.

Key	Value
Kind	Trust Boundary
Name	AWS Control Boundary
Mentioned In	Security View

TB-BEDROCK-GITHUB-EXTERNAL

Boundary between external GitHub traffic and internal review processing.

Key	Value
Kind	Trust Boundary
Name	GitHub External Boundary
Mentioned In	n/a

14.2. Catalog References

ACT-DEVELOPER

Engineer opening and updating pull requests.

Key	Value
Kind	Catalog Actor
Name	developer
Mentioned In	REQ-PRR-002

ACT-PLATFORM-OPERATOR

Owner of deployment/runtime operations and reliability controls.

Key	Value
Kind	Catalog Actor
Name	platform operator
Mentioned In	REQ-PRR-010

ACT-REPOSITORY-MAINTAINER

Owner of merge decisions and repository quality gates.

Key	Value
Kind	Catalog Actor
Name	repository maintainer
Mentioned In	REQ-PRR-005 , REQ-PRR-009 , REQ-PRR-010

ACT-SECURITY-ENGINEER

Owner of security policy checks and finding triage.

Key	Value
Kind	Catalog Actor
Name	security engineer
Mentioned In	Security View , REQ-PRR-007

ai review enabled

Feature flag enabling model-assisted review.

Key	Value
Kind	Catalog Alias
Name	ai review enabled
Mentioned In	n/a

audit enabled

Feature flag enabling audit metadata persistence.

Key	Value
Kind	Catalog Alias
Name	audit enabled
Mentioned In	n/a

authorized events

Event raised when new commits are pushed to an open pull request.

Key	Value
Kind	Catalog Alias

Key	Value
Name	authorized events
Mentioned In	n/a

aws sdk

SDK dependency used for AWS runtime integrations.

Key	Value
Kind	Catalog Alias
Name	aws sdk
Mentioned In	n/a

bedrock api

External Bedrock runtime endpoint used for model inference.

Key	Value
Kind	Catalog Alias
Name	bedrock api
Mentioned In	n/a

bedrock unavailable

Event raised when Bedrock inference path is degraded.

Key	Value
Kind	Catalog Alias
Name	bedrock unavailable
Mentioned In	n/a

check run summaries

Aggregated review result posted as a GitHub check run.

Key	Value
Kind	Catalog Alias
Name	check run summaries

Key	Value
Mentioned In	n/a

context assembly

Fetches changed files and prepares bounded review context payloads.

Key	Value
Kind	Catalog Alias
Name	context assembly
Mentioned In	Architecture Intent View , Security View

context prepared

Event indicating context assembly completed and is ready for analysis.

Key	Value
Kind	Catalog Alias
Name	context prepared
Mentioned In	n/a

deterministic checks

Evaluates deterministic repository policy checks.

Key	Value
Kind	Catalog Alias
Name	deterministic checks
Mentioned In	Architecture Intent View

findings generated

Event indicating analysis output is ready for publication.

Key	Value
Kind	Catalog Alias
Name	findings generated
Mentioned In	n/a

findings published

Event indicating review output has been published to GitHub checks/comments.

Key	Value
Kind	Catalog Alias
Name	findings published
Mentioned In	n/a

github api

External GitHub API used for pull request metadata and review publication.

Key	Value
Kind	Catalog Alias
Name	github api
Mentioned In	REQ-PRR-002

github integration

Functional group for webhook ingress and pull request context intake.

Key	Value
Kind	Catalog Alias
Name	github integration
Mentioned In	n/a

github sdk

SDK dependency used for GitHub App REST interactions.

Key	Value
Kind	Catalog Alias
Name	github sdk
Mentioned In	n/a

lambda ops

Manages runtime deployment and operational controls.

Key	Value
Kind	Catalog Alias
Name	lambda ops
Mentioned In	n/a

maintainer

Owner of merge decisions and repository quality gates.

Key	Value
Kind	Catalog Alias
Name	maintainer
Mentioned In	n/a

normal mode

Default operating mode using AI plus deterministic checks.

Key	Value
Kind	Catalog Alias
Name	normal mode
Mentioned In	n/a

normalized review context

Pull request file changes and metadata prepared for analysis.

Key	Value
Kind	Catalog Alias
Name	normalized review context
Mentioned In	n/a

orchestration

Coordinates Bedrock analysis with deterministic checks and fallback logic.

Key	Value
Kind	Catalog Alias

Key	Value
Name	orchestration
Mentioned In	Architecture Intent View , Deployment View

platform ops

Owner of deployment/runtime operations and reliability controls.

Key	Value
Kind	Catalog Alias
Name	platform ops
Mentioned In	n/a

policy bundle

Deterministic review policy bundle used during analysis.

Key	Value
Kind	Catalog Alias
Name	policy bundle
Mentioned In	n/a

policy checks enabled

Feature flag enabling deterministic policy checks.

Key	Value
Kind	Catalog Alias
Name	policy checks enabled
Mentioned In	n/a

pr author

Engineer opening and updating pull requests.

Key	Value
Kind	Catalog Alias
Name	pr author

Key	Value
Mentioned In	n/a

pr opened

Event raised when a new pull request is opened.

Key	Value
Kind	Catalog Alias
Name	pr opened
Mentioned In	n/a

pr spam

High-volume event abuse causing review capacity exhaustion.

Key	Value
Kind	Catalog Alias
Name	pr spam
Mentioned In	n/a

pr synchronize

Event raised when new commits are pushed to an open pull request.

Key	Value
Kind	Catalog Alias
Name	pr synchronize
Mentioned In	n/a

prompt injection

Malicious prompt-like content embedded in pull request code changes.

Key	Value
Kind	Catalog Alias
Name	prompt injection
Mentioned In	Security View

publication

Publishes review findings to GitHub checks and comments.

Key	Value
Kind	Catalog Alias
Name	publication
Mentioned In	Architecture Intent View , Deployment View , Security View , REQ-PRR-005 , REQ-PRR-009

publication output

Event indicating review output has been published to GitHub checks/comments.

Key	Value
Kind	Catalog Alias
Name	publication output
Mentioned In	n/a

pull request files

Pull request file changes and metadata prepared for analysis.

Key	Value
Kind	Catalog Alias
Name	pull request files
Mentioned In	n/a

pull request metadata

Pull request file changes and metadata prepared for analysis.

Key	Value
Kind	Catalog Alias
Name	pull request metadata
Mentioned In	n/a

rate limit exceeded

Event raised when PR review rate guardrails are hit.

Key	Value
Kind	Catalog Alias
Name	rate limit exceeded
Mentioned In	n/a

review active

State where review processing is currently active for a pull request.

Key	Value
Kind	Catalog Alias
Name	review active
Mentioned In	n/a

review context

Pull request file changes and metadata prepared for analysis.

Key	Value
Kind	Catalog Alias
Name	review context
Mentioned In	Architecture Intent View

review outcome

Event indicating analysis output is ready for publication.

Key	Value
Kind	Catalog Alias
Name	review outcome
Mentioned In	REQ-PRR-006

review system

Functional group for AI and deterministic review analysis and publication.

Key	Value
Kind	Catalog Alias

Key	Value
Name	review system
Mentioned In	n/a

review-ready context

Pull request file changes and metadata prepared for analysis.

Key	Value
Kind	Catalog Alias
Name	review-ready context
Mentioned In	n/a

runtime operations

Functional group for deployment lifecycle and secret/config controls.

Key	Value
Kind	Catalog Alias
Name	runtime operations
Mentioned In	Architecture Intent View

secret config

Manages credentials and secure runtime configuration.

Key	Value
Kind	Catalog Alias
Name	secret config
Mentioned In	n/a

secret leakage

Unintended credential disclosure in generated review findings.

Key	Value
Kind	Catalog Alias
Name	secret leakage

Key	Value
Mentioned In	n/a

secrets manager

Managed secret store for application credentials and config values.

Key	Value
Kind	Catalog Alias
Name	secrets manager
Mentioned In	n/a

security reviewer

Owner of security policy checks and finding triage.

Key	Value
Kind	Catalog Alias
Name	security reviewer
Mentioned In	n/a

signature valid

Condition indicating webhook HMAC verification succeeded.

Key	Value
Kind	Catalog Alias
Name	signature valid
Mentioned In	n/a

the pr review system

GitHub App review system-of-interest that ingests PR events and publishes review outcomes.

Key	Value
Kind	Catalog Alias
Name	the pr review system

Key	Value
Mentioned In	REQ-PRR-001 , REQ-PRR-002 , REQ-PRR-003 , REQ-PRR-004 , REQ-PRR-005 , REQ-PRR-006 , REQ-PRR-007 , REQ-PRR-008 , REQ-PRR-009 , REQ-PRR-010

webhook events

Event raised when a new pull request is opened.

Key	Value
Kind	Catalog Alias
Name	webhook events
Mentioned In	n/a

webhook ingress

Validates webhook signatures and routes trusted pull request events.

Key	Value
Kind	Catalog Alias
Name	webhook ingress
Mentioned In	Deployment View

webhook spoofing

Forged webhook request attempting unauthorized review execution.

Key	Value
Kind	Catalog Alias
Name	webhook spoofing
Mentioned In	n/a

AV-PR-SPAM-ABUSE

High-volume event abuse causing review capacity exhaustion.

Key	Value
Kind	Catalog Attack Vector
Name	pull request spam abuse

Key	Value
Mentioned In	n/a

AV-PROMPT-INJECTION-IN-DIFF

Malicious prompt-like content embedded in pull request code changes.

Key	Value
Kind	Catalog Attack Vector
Name	prompt injection in diff
Mentioned In	Security View

AV-SECRET-LEAKAGE-IN-REVIEW

Unintended credential disclosure in generated review findings.

Key	Value
Kind	Catalog Attack Vector
Name	secret leakage in review output
Mentioned In	Security View

AV-SPOOFED-GITHUB-WEBHOOK

Forged webhook request attempting unauthorized review execution.

Key	Value
Kind	Catalog Attack Vector
Name	spoofed github webhook
Mentioned In	Security View

COND-WEBHOOK-SIGNATURE-VALID

Condition indicating webhook HMAC verification succeeded.

Key	Value
Kind	Catalog Condition
Name	webhook signature is valid
Mentioned In	REQ-PRR-001

DATA-CHECKRUN-SUMMARY

Aggregated review result posted as a GitHub check run.

Key	Value
Kind	Catalog Data Term
Name	check run summary
Mentioned In	REQ-PRR-005

DATA-DIFF-CONTEXT

Pull request file changes and metadata prepared for analysis.

Key	Value
Kind	Catalog Data Term
Name	diff context
Mentioned In	REQ-PRR-002

DATA-REDACTED-AUDIT-METADATA

Sanitized prompt and finding metadata retained for audit.

Key	Value
Kind	Catalog Data Term
Name	redacted audit metadata
Mentioned In	REQ-PRR-007

DATA-RESULT-METADATA

Metadata describing review output characteristics for audit and reporting.

Key	Value
Kind	Catalog Data Term
Name	result metadata
Mentioned In	REQ-PRR-009

DATA-WEBHOOK-SIGNATURE

Signature used to verify webhook authenticity.

Key	Value
Kind	Catalog Data Term
Name	webhook signature
Mentioned In	Architecture Intent View , Security View , REQ-PRR-001

EVT-BEDROCK-UNAVAILABLE

Event raised when Bedrock inference path is degraded.

Key	Value
Kind	Catalog Event
Name	bedrock runtime is unavailable
Mentioned In	REQ-PRR-006

EVT-CONTEXT-PREPARED

Event indicating context assembly completed and is ready for analysis.

Key	Value
Kind	Catalog Event
Name	diff context is prepared
Mentioned In	REQ-PRR-003

EVT-FINDINGS-GENERATED

Event indicating analysis output is ready for publication.

Key	Value
Kind	Catalog Event
Name	review findings are generated
Mentioned In	REQ-PRR-005 , REQ-PRR-009

EVT-FINDINGS-PUBLISHED

Event indicating review output has been published to GitHub checks/comments.

Key	Value
Kind	Catalog Event

Key	Value
Name	review findings are published
Mentioned In	REQ-PRR-007

EVT-PR-OPENED

Event raised when a new pull request is opened.

Key	Value
Kind	Catalog Event
Name	pull request opened event is received
Mentioned In	REQ-PRR-002

EVT-PR-SYNCHRONIZE

Event raised when new commits are pushed to an open pull request.

Key	Value
Kind	Catalog Event
Name	pull request synchronize event is received
Mentioned In	REQ-PRR-001

EVT-RATE-LIMIT-EXCEEDED

Event raised when PR review rate guardrails are hit.

Key	Value
Kind	Catalog Event
Name	review request rate limit is exceeded
Mentioned In	REQ-PRR-008 , REQ-PRR-010

FEAT-AI-REVIEW

Feature flag enabling model-assisted review.

Key	Value
Kind	Catalog Feature
Name	ai review is enabled

Key	Value
Mentioned In	REQ-PRR-003

FEAT-AUDIT-LOGGING

Feature flag enabling audit metadata persistence.

Key	Value
Kind	Catalog Feature
Name	audit logging is enabled
Mentioned In	REQ-PRR-007

FEAT-POLICY-CHECKS

Feature flag enabling deterministic policy checks.

Key	Value
Kind	Catalog Feature
Name	policy checks are enabled
Mentioned In	REQ-PRR-004

FG-PLATFORM-OPERATIONS

Functional group for deployment lifecycle and secret/config controls.

Key	Value
Kind	Catalog Functional Group
Name	platform operations
Mentioned In	Architecture Intent View , Deployment View

FG-PR-INTEGRATION

Functional group for webhook ingress and pull request context intake.

Key	Value
Kind	Catalog Functional Group
Name	pr integration
Mentioned In	Architecture Intent View , Deployment View

FG-REVIEW-INTELLIGENCE

Functional group for AI and deterministic review analysis and publication.

Key	Value
Kind	Catalog Functional Group
Name	review intelligence
Mentioned In	Architecture Intent View , Deployment View

FU-GITHUB-WEBHOOK-INGRESS

Validates webhook signatures and routes trusted pull request events.

Key	Value
Kind	Catalog Functional Unit
Name	github webhook ingress
Mentioned In	Architecture Intent View , Security View , Pr Opened Review Flow , Rate Limit Deferral Test

FU-LAMBDA-RUNTIME-OPERATIONS

Manages runtime deployment and operational controls.

Key	Value
Kind	Catalog Functional Unit
Name	lambda runtime operations
Mentioned In	Architecture Intent View , Communication View , Deployment View , Security View , Traceability View

FU-POLICY-CHECKS

Evaluates deterministic repository policy checks.

Key	Value
Kind	Catalog Functional Unit
Name	policy checks
Mentioned In	Architecture Intent View , Communication View , Deployment View , Security View , Traceability View , Interaction Flow View , REQ-PRR-004 , Policy Only Fallback Test

FU-PR-CONTEXT-ASSEMBLY

Fetches changed files and prepares bounded review context payloads.

Key	Value
Kind	Catalog Functional Unit
Name	pr context assembly
Mentioned In	Architecture Intent View , Communication View , Deployment View , Security View , Traceability View , Pr Opened Review Flow

FU-REVIEW-ORCHESTRATION

Coordinates Bedrock analysis with deterministic checks and fallback logic.

Key	Value
Kind	Catalog Functional Unit
Name	review orchestration
Mentioned In	Architecture Intent View , Deployment View , Security View , Pr Opened Review Flow , Policy Only Fallback Test

FU-REVIEW-PUBLICATION

Publishes review findings to GitHub checks and comments.

Key	Value
Kind	Catalog Functional Unit
Name	review publication
Mentioned In	Architecture Intent View , Deployment View , Security View , Pr Opened Review Flow , Audit Redaction Test , Policy Only Fallback Test , Github Check Run Contract Test , Rate Limit Deferral Test

FU-SECRETS-CONFIGURATION

Manages credentials and secure runtime configuration.

Key	Value
Kind	Catalog Functional Unit
Name	secrets and configuration
Mentioned In	Architecture Intent View , Communication View , Deployment View , Security View , Traceability View , Audit Redaction Test

MODE-NORMAL

Default operating mode using AI plus deterministic checks.

Key	Value
Kind	Catalog Mode
Name	normal mode is active
Mentioned In	REQ-PRR-001

REF-AWS-BEDROCK-RUNTIME

External Bedrock runtime endpoint used for model inference.

Key	Value
Kind	Catalog Referenced Element
Name	bedrock runtime
Mentioned In	Architecture Intent View , Deployment View , Security View , REQ-PRR-003

REF-AWS-SDK-GO

SDK dependency used for AWS runtime integrations.

Key	Value
Kind	Catalog Referenced Element
Name	aws sdk for go
Mentioned In	Architecture Intent View , Deployment View , Security View

REF-AWS-SECRETS-MANAGER

Managed secret store for application credentials and config values.

Key	Value
Kind	Catalog Referenced Element
Name	aws secrets manager
Mentioned In	Architecture Intent View , Deployment View , Security View

REF-GITHUB-APP-API

External GitHub API used for pull request metadata and review publication.

Key	Value
Kind	Catalog Referenced Element
Name	github app api
Mentioned In	Architecture Intent View , Deployment View , Security View

REF-GITHUB-REST-SDK

SDK dependency used for GitHub App REST interactions.

Key	Value
Kind	Catalog Referenced Element
Name	github rest sdk
Mentioned In	Architecture Intent View , Deployment View , Security View

REF-POLICY-RULESET

Deterministic review policy bundle used during analysis.

Key	Value
Kind	Catalog Referenced Element
Name	policy ruleset
Mentioned In	Architecture Intent View , Deployment View

STATE-PR-REVIEW-ACTIVE

State where review processing is currently active for a pull request.

Key	Value
Kind	Catalog State
Name	pull request review is active
Mentioned In	REQ-PRR-002 , REQ-PRR-004 , REQ-PRR-005 , REQ-PRR-009

SYS-PR-REVIEW-SYSTEM

GitHub App review system-of-interest that ingests PR events and publishes review outcomes.

Key	Value
Kind	Catalog System

Key	Value
Name	pr review system
Mentioned In	n/a

EM-ACTOR

A person or operational role that interacts with functional units.

Key	Value
Kind	Engineering Model Term
Name	actor
Mentioned In	n/a

EM-ATTACK-VECTOR

A technical misuse or attack path that targets functional, referenced, or runtime elements.

Key	Value
Kind	Engineering Model Term
Name	attack vector
Mentioned In	n/a

EM-AUTHORED-MAPPING

An explicit relationship declared in architecture inputs between authored or referenced elements.

Key	Value
Kind	Engineering Model Term
Name	authored mapping
Mentioned In	Communication View , Security View

EM-CODE-ELEMENT

An inferred code structure or ownership element discovered from source trees and build metadata.

Key	Value
Kind	Engineering Model Term
Name	code element

Key	Value
Mentioned In	n/a

EM-CONTROL

An authored security, policy, or operational control used to constrain behavior and reduce risk.

Key	Value
Kind	Engineering Model Term
Name	control
Mentioned In	Deployment View , Security View

EM-DATA-OBJECT

An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.

Key	Value
Kind	Engineering Model Term
Name	data object
Mentioned In	n/a

EM-DEPLOYMENT-TARGET

An authored deployment destination such as environment, cluster, namespace, or registry scope.

Key	Value
Kind	Engineering Model Term
Name	deployment target
Mentioned In	n/a

EM-EVENT

An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.

Key	Value
Kind	Engineering Model Term
Name	event
Mentioned In	Architecture Intent View , Communication View , Security View , State Lifecycle View

EM-FLOW

An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.

Key	Value
Kind	Engineering Model Term
Name	flow
Mentioned In	Architecture Intent View , Communication View , Deployment View , Interaction Flow View , Pr Opened Review Flow

EM-FLOW-STEP

A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.

Key	Value
Kind	Engineering Model Term
Name	flow step
Mentioned In	n/a

EM-FUNCTIONAL-GROUP

A major authored capability area that groups related functional units.

Key	Value
Kind	Engineering Model Term
Name	functional group
Mentioned In	n/a

EM-FUNCTIONAL-UNIT

An authored working unit inside a functional group that owns specific behavior.

Key	Value
Kind	Engineering Model Term
Name	functional unit
Mentioned In	State Lifecycle View

EM-INFERRED-MAPPING

A discovered relationship that links inferred runtime/code elements upward to authored design.

Key	Value
Kind	Engineering Model Term
Name	inferred mapping
Mentioned In	n/a

EM-INTERFACE

An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.

Key	Value
Kind	Engineering Model Term
Name	interface
Mentioned In	Communication View

EM-REFERENCED-ELEMENT

An architecture-relevant external, platform, or third-party dependency represented by role.

Key	Value
Kind	Engineering Model Term
Name	referenced element
Mentioned In	n/a

EM-REQUIREMENT

A structured requirement statement that defines expected system behavior and maps to functional ownership.

Key	Value
Kind	Engineering Model Term
Name	requirement
Mentioned In	Architecture Intent View , Deployment View , Traceability View , Interaction Flow View , REQ-PRR-006

EM-RUNTIME-ELEMENT

An inferred runtime realization element discovered from infrastructure and deployment sources.

Key	Value
Kind	Engineering Model Term
Name	runtime element
Mentioned In	n/a

EM-STATE

An authored lifecycle state used to model transition behavior, guards, and event-driven progression.

Key	Value
Kind	Engineering Model Term
Name	state
Mentioned In	State Lifecycle View

EM-TRUST-BOUNDARY

An authored trust separation zone that marks policy, identity, or security control boundaries.

Key	Value
Kind	Engineering Model Term
Name	trust boundary
Mentioned In	Architecture Intent View , Security View , REQ-PRR-001

EM-UPWARD-LINKING

Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.

Key	Value
Kind	Engineering Model Term
Name	upward linking
Mentioned In	n/a

EM-VERIFICATION-CHECK

An inferred or authored verification artifact that validates requirement behavior with test evidence.

Key	Value
Kind	Engineering Model Term
Name	verification check
Mentioned In	n/a

14.3. Inferred Runtime References

review_orchestrator

assembles diff context and orchestrates deterministic plus Bedrock-backed review analysis

Key	Value
Kind	Runtime lambda_function
Owner	FU-REVIEW-ORCHESTRATION
Source	examples/bedrock-pr-review-github-app-sample/infra/terraform/main.tf
Mentioned In	Deployment View , Security View

review_publisher

publishes check run summaries and inline review comments back to GitHub

Key	Value
Kind	Runtime lambda_function
Owner	FU-REVIEW-PUBLICATION
Source	examples/bedrock-pr-review-github-app-sample/infra/terraform/main.tf
Mentioned In	Deployment View , Security View

webhook_ingress

validates GitHub webhook signature and normalizes pull request event payloads

Key	Value
Kind	Runtime lambda_function
Owner	FU-GITHUB-WEBHOOK-INGRESS
Source	examples/bedrock-pr-review-github-app-sample/infra/terraform/main.tf
Mentioned In	Deployment View , Security View

14.4. Inferred Code References

aws-sdk-go-v2/secretsmanager (secrets_configuration.ts)

Inferred library_external dependency owned by FU-SECRETS-CONFIGURATION from secrets_configuration.ts.

Key	Value
Kind	Code library_external
Owner	FU-SECRETS-CONFIGURATION
Source	secrets_configuration.ts
Mentioned In	n/a

github-rest-sdk (pr_context_assembly.ts)

Inferred library_external dependency owned by FU-PR-CONTEXT-ASSEMBLY from pr_context_assembly.ts.

Key	Value
Kind	Code library_external
Owner	FU-PR-CONTEXT-ASSEMBLY
Source	pr_context_assembly.ts
Mentioned In	n/a

github-rest-sdk/publication-client (review_publication.ts)

Inferred library_external dependency owned by FU-REVIEW-PUBLICATION from review_publication.ts.

Key	Value
Kind	Code library_external
Owner	FU-REVIEW-PUBLICATION
Source	review_publication.ts
Mentioned In	n/a

serde_json::json (review_orchestrator.rs)

Inferred library_external dependency owned by FU-REVIEW-ORCHESTRATION from review_orchestrator.rs.

Key	Value
Kind	Code library_external

Key	Value
Owner	FU-REVIEW-ORCHESTRATION
Source	review_orchestrator.rs
Mentioned In	n/a

crypto/hmac (webhook_ingress.go)

Inferred library_stdlib dependency owned by FU-GITHUB-WEBHOOK-INGRESS from webhook_ingress.go.

Key	Value
Kind	Code library_stdlib
Owner	FU-GITHUB-WEBHOOK-INGRESS
Source	webhook_ingress.go
Mentioned In	n/a

crypto/sha256 (webhook_ingress.go)

Inferred library_stdlib dependency owned by FU-GITHUB-WEBHOOK-INGRESS from webhook_ingress.go.

Key	Value
Kind	Code library_stdlib
Owner	FU-GITHUB-WEBHOOK-INGRESS
Source	webhook_ingress.go
Mentioned In	n/a

encoding/hex (webhook_ingress.go)

Inferred library_stdlib dependency owned by FU-GITHUB-WEBHOOK-INGRESS from webhook_ingress.go.

Key	Value
Kind	Code library_stdlib
Owner	FU-GITHUB-WEBHOOK-INGRESS
Source	webhook_ingress.go
Mentioned In	n/a

fmt (webhook_ingress.go)

Inferred library_stdlib dependency owned by FU-GITHUB-WEBHOOK-INGRESS from webhook_ingress.go.

Key	Value
Kind	Code library_stdlib
Owner	FU-GITHUB-WEBHOOK-INGRESS
Source	webhook_ingress.go
Mentioned In	n/a

log (lambda_runtime_ops.go)

Inferred library_stdlib dependency owned by FU-LAMBDA-RUNTIME-OPERATIONS from lambda_runtime_ops.go.

Key	Value
Kind	Code library_stdlib
Owner	FU-LAMBDA-RUNTIME-OPERATIONS
Source	lambda_runtime_ops.go
Mentioned In	n/a

lambda_runtime_ops.go

wires Lambda runtime entrypoints and shared execution dependencies

Key	Value
Kind	Code source_file
Owner	FU-LAMBDA-RUNTIME-OPERATIONS
Source	lambda_runtime_ops.go
Mentioned In	Deployment View

policy_checks.rs

evaluates policy and style rules before review publication

Key	Value
Kind	Code source_file
Owner	FU-POLICY-CHECKS
Source	policy_checks.rs

Key	Value
Mentioned In	Deployment View

pr_context_assembly.ts

collects pull request diff context and prepares review input payloads

Key	Value
Kind	Code source_file
Owner	FU-PR-CONTEXT-ASSEMBLY
Source	pr_context_assembly.ts
Mentioned In	Deployment View , Security View

review_orchestrator.rs

coordinates deterministic and AI review steps for pull requests

Key	Value
Kind	Code source_file
Owner	FU-REVIEW-ORCHESTRATION
Source	review_orchestrator.rs
Mentioned In	n/a

review_publication.ts

publishes review comments and check-run outcomes back to GitHub

Key	Value
Kind	Code source_file
Owner	FU-REVIEW-PUBLICATION
Source	review_publication.ts
Mentioned In	Deployment View , Security View

secrets_configuration.ts

loads GitHub and Bedrock secret configuration for review execution

Key	Value
Kind	Code source_file
Owner	FU-SECRETS-CONFIGURATION
Source	secrets_configuration.ts
Mentioned In	Deployment View , Security View

webhook_ingress.go

validates GitHub webhook signatures and routes pull request events

Key	Value
Kind	Code source_file
Owner	FU-GITHUB-WEBHOOK-INGRESS
Source	webhook_ingress.go
Mentioned In	n/a

lambda_runtime_ops.go:10

Inferred code symbol owned by FU-LAMBDA-RUNTIME-OPERATIONS from lambda_runtime_ops.go:10.

Key	Value
Kind	Code symbol
Owner	FU-LAMBDA-RUNTIME-OPERATIONS
Source	lambda_runtime_ops.go:10
Mentioned In	n/a

policy_checks.rs:21

Inferred code symbol owned by FU-POLICY-CHECKS from policy_checks.rs:21.

Key	Value
Kind	Code symbol
Owner	FU-POLICY-CHECKS
Source	policy_checks.rs:21
Mentioned In	n/a

policy_checks.rs:7

Inferred code symbol owned by FU-POLICY-CHECKS from policy_checks.rs:7.

Key	Value
Kind	Code symbol
Owner	FU-POLICY-CHECKS
Source	policy_checks.rs:7
Mentioned In	n/a

pr_context_assembly.ts:10

Inferred code symbol owned by FU-PR-CONTEXT-ASSEMBLY from pr_context_assembly.ts:10.

Key	Value
Kind	Code symbol
Owner	FU-PR-CONTEXT-ASSEMBLY
Source	pr_context_assembly.ts:10
Mentioned In	n/a

pr_context_assembly.ts:16

Inferred code symbol owned by FU-PR-CONTEXT-ASSEMBLY from pr_context_assembly.ts:16.

Key	Value
Kind	Code symbol
Owner	FU-PR-CONTEXT-ASSEMBLY
Source	pr_context_assembly.ts:16
Mentioned In	n/a

pr_context_assembly.ts:7

Inferred code symbol owned by FU-PR-CONTEXT-ASSEMBLY from pr_context_assembly.ts:7.

Key	Value
Kind	Code symbol
Owner	FU-PR-CONTEXT-ASSEMBLY

Key	Value
Source	pr_context_assembly.ts:7
Mentioned In	n/a

review_orchestrator.rs:19

Inferred code symbol owned by FU-REVIEW-ORCHESTRATION from review_orchestrator.rs:19.

Key	Value
Kind	Code symbol
Owner	FU-REVIEW-ORCHESTRATION
Source	review_orchestrator.rs:19
Mentioned In	n/a

review_orchestrator.rs:9

Inferred code symbol owned by FU-REVIEW-ORCHESTRATION from review_orchestrator.rs:9.

Key	Value
Kind	Code symbol
Owner	FU-REVIEW-ORCHESTRATION
Source	review_orchestrator.rs:9
Mentioned In	n/a

review_publication.ts:10

Inferred code symbol owned by FU-REVIEW-PUBLICATION from review_publication.ts:10.

Key	Value
Kind	Code symbol
Owner	FU-REVIEW-PUBLICATION
Source	review_publication.ts:10
Mentioned In	n/a

review_publication.ts:23

Inferred code symbol owned by FU-REVIEW-PUBLICATION from review_publication.ts:23.

Key	Value
Kind	Code symbol
Owner	FU-REVIEW-PUBLICATION
Source	review_publication.ts:23
Mentioned In	n/a

review_publication.ts:28

Inferred code symbol owned by FU-REVIEW-PUBLICATION from review_publication.ts:28.

Key	Value
Kind	Code symbol
Owner	FU-REVIEW-PUBLICATION
Source	review_publication.ts:28
Mentioned In	n/a

review_publication.ts:7

Inferred code symbol owned by FU-REVIEW-PUBLICATION from review_publication.ts:7.

Key	Value
Kind	Code symbol
Owner	FU-REVIEW-PUBLICATION
Source	review_publication.ts:7
Mentioned In	n/a

secrets_configuration.ts:10

Inferred code symbol owned by FU-SECRETS-CONFIGURATION from secrets_configuration.ts:10.

Key	Value
Kind	Code symbol
Owner	FU-SECRETS-CONFIGURATION
Source	secrets_configuration.ts:10
Mentioned In	n/a

secrets_configuration.ts:7

Inferred code symbol owned by FU-SECRETS-CONFIGURATION from secrets_configuration.ts:7.

Key	Value
Kind	Code symbol
Owner	FU-SECRETS-CONFIGURATION
Source	secrets_configuration.ts:7
Mentioned In	n/a

webhook_ingress.go:15

Inferred code symbol owned by FU-GITHUB-WEBHOOK-INGRESS from webhook_ingress.go:15.

Key	Value
Kind	Code symbol
Owner	FU-GITHUB-WEBHOOK-INGRESS
Source	webhook_ingress.go:15
Mentioned In	n/a

webhook_ingress.go:25

Inferred code symbol owned by FU-GITHUB-WEBHOOK-INGRESS from webhook_ingress.go:25.

Key	Value
Kind	Code symbol
Owner	FU-GITHUB-WEBHOOK-INGRESS
Source	webhook_ingress.go:25
Mentioned In	n/a

14.5. Verification References

VER-INF-E2E-PR_OPENED_REVIEW_FLOW-6BF0B5

exercises end-to-end pull request opened flow from webhook ingest to published review

Key	Value
Kind	e2e

Key	Value
Status	pass
Source	test-results/contract-checkrun.json (+2)
Name	Pr Opened Review Flow
Mentioned In	Interaction Flow View , Pr Opened Review Flow

VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-B108A1

checks audit evidence redaction for secrets and sensitive payload fields

Key	Value
Kind	integration
Status	not-run
Source	tests/integration/audit_redaction_test.ts
Name	Audit Redaction Test
Mentioned In	Interaction Flow View , Audit Redaction Test

VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-76E20A

verifies policy-only fallback when Bedrock inference is unavailable

Key	Value
Kind	integration
Status	pass
Source	test-results/integration-policy-fallback.json (+1)
Name	Policy Only Fallback Test
Mentioned In	Interaction Flow View , Policy Only Fallback Test

VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-17DBA6

validates GitHub check-run contract shape and required review result fields

Key	Value
Kind	test
Status	not-run

Key	Value
Source	tests/contract/github_check_run_contract_test.go
Name	Github Check Run Contract Test
Mentioned In	Interaction Flow View , Github Check Run Contract Test

VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-DC9EB2

verifies rate-limit deferral behavior for pull request review processing

Key	Value
Kind	unit
Status	pass
Source	test-results/unit-rate-limit.xml (+1)
Name	Rate Limit Deferral Test
Mentioned In	Interaction Flow View , Rate Limit Deferral Test